



Amanzi Water Ordering Information System: A Digital Solution for Efficient, Secure, and Structured Water Delivery at North-West University, Mahikeng Campus.



**AUTHORS: BP MOGOLA 40877825, K MAKAU 33672040, TP
SETHWAPE 42501512**

LIST OF ABBREVIATIONS

NWU.....	North-West University
CSS.....	Cascading Style Sheet
DFD.....	Data Flow Diagram
API.....	Application Programming Interface
WBS.....	Work Breakdown Structure
DB.....	Database
ERD.....	Entity Relationship Diagram

TABLE OF CONTENTS

LIST OF ABBREVIATIONS	I
PHASE 1 – SYSTEMS PLANNING.....	1
1 PRELIMINARY INVESTIGATION REPORT.....	1
1.1 Step 1 – Understand the problem or opportunity.....	1
1.2 Step 5 – Evaluate feasibility.....	18
1.2.1 Operational feasibility	18
1.2.2 Technical feasibility.....	18
1.2.3 Economic feasibility	18
1.2.4 Schedule feasibility	18
1.2.4.1 Sampling technique (Heading 4 style)	19
PHASE 2 – SYSTEMS ANALYSIS	20
2 SYSTEM REQUIREMENTS MODELING	20
2.1 SYSTEM REQUIREMENTS	20
2.1.1 Output	21
2.1.2 Security	21
2.2 DATA AND PROCESS MODELING	22-33
PHASE 3 - SYSTEMS DESIGN	33
3 SYSTEM DESIGN SPECIFICATION.....	33-50
PHASE 4 – SYSTEMS IMPLEMENTATION	51

4	A FUNCTIONING INFORMATION SYSTEM.....	51-58
	PHASE 5 – SYSTEMS SUPPORT AND SECURITY	59
5	AN OPERATIONAL SYSTEM THAT IS PROPERLY MAINTAINED, SUPPORTED, AND SECURED	59
	BIBLIOGRAPHY:	63
	ANNEXURE A: DATA COLLECTION TOOL.....	64
	ANNEXURE B: SYSTEM PERFORMANCE EVALUATION TOOL.....	65
	ANNEXURE C: SYSTEM SECURITY AUDIT TOOL.....	66

LIST OF TABLES

Table 1:	Work Breakdown Structure - WBS	19
Table 2:	STACKHOLDER'S LIST.....	20
Table 3:	New Customers Sign up case description.....	25
Table 4:	Customers login use case description.....	26
Table 5:	Order case description.....	27
Table 6:	View Order case description.....	28
Table 7:	View Statistics.....	29
Table 8:	New Service Provider Sign up case description.....	30
Table 9:	Login case description for service providers.....	31
Table 10:	View Orders Case description.....	31
Table 11:	View Historical Orders.....	32
Table 12:	View Statistics.....	32
Table 13:	Test Case 1 - Register New Student.....	51
Table 14:	Test Case 2 - Login Student.....	52
Table 15:	Test Case 3 - Order.....	53
Table 16:	Test Case 4 - View Status.....	54
Table 17:	Test Case 5 - View Order History.....	54
Table 18:	Test Case 6 - Register New Service provider.....	55
Table 19:	Test Case 7 - Login.....	56
Table 20:	Test Case 8 - View Order History.....	57
Table 21:	Test Case 9 - View Order Analytics.....	57

LIST OF FIGURES

Figure 1-30:	Fact Findings	17-17
Figure 31:	Fishbone diagram	17
Figure 32:	Work-Breakdown-Structure.....	19
Figure 33:	DFD Student/Customer.....	22
Figure 34:	DFD Supplier.....	23
Figure 35:	Use case diagram.....	24
Figure 36:	Software Architecture.....	34
Figure 37:	Welcome Home Page (GUI).....	35
Figure 38:	Register Form (GUI).....	35
Figure 39:	Login form (GUI).....	36
Figure 40:	Forgot Password (GUI).....	36
Figure 41:	Reset Password email (GUI).....	37
Figure 42:	Reset Password Form (GUI).....	37
Figure 43:	Customer Profile (GUI).....	38
Figure 44:	Customer Dashboard (GUI).....	38
Figure 45:	Customer Dashboard Cont...(GUI).....	39
Figure 46:	Customer Pending Orders (GUI).....	39
Figure 47:	Customer Order History (GUI).....	40
Figure 48:	Customer Completed Orders (GUI).....	40
Figure 49:	Customer Order Details/Invoice (GUI).....	41
Figure 50:	Supplier Profile (GUI).....	41

Figure 51:	Supplier Dashboard (GUI).....	42
Figure 52:	Supplier Order Management (GUI).....	42
Figure 53:	Supplier Availability Management (GUI).....	43
Figure 54:	Supplier Sales Analysis (GUI).....	43
Figure 55:	Entity Relationship Diagram.....	44
Figure 56:	Class Diagram.....	45
Figure 57:	Component Diagram.....	46
Figure 58:	Deployment Diagram.....	46
Figure 59:	Student Sequence Diagram (Place Order)	47
Figure 60:	Gantt Chart.....	51

PHASE 1 – SYSTEMS PLANNING

1 PRELIMINARY INVESTIGATION REPORT

This section provides an overview of the existing problems faced in the current water ordering process at North-West University, Mahikeng Campus. It highlights the main challenges, such as lack of privacy on WhatsApp, inconsistent delivery times, and the absence of proper tracking for orders, spending, and income. The purpose of this investigation is to understand these issues, identify user needs, and determine whether developing a digital water ordering system (Amanzi) would be a practical and effective solution.

1.1 Step 1 – Understand the problem or opportunity

Students residing at the North-West University, Mahikeng campus currently rely on manual ordering method for purified water, which lacks a convenient digital platform for students residing on-campus to order purified water for delivery. Both water suppliers and students rely mainly on WhatsApp and Phone calls to communicate and place orders, which lacks confidentiality/privacy for students who do not wish for their contact details to be known by strangers in a case whereby the suppliers add them into the WhatsApp groups for ordering. Many of these water suppliers do not have a structured sales tracking system and there isn't a structured way of tracking monthly spending for students. Furthermore, there is no structured scheduling of deliveries. The Amanzi Information System aims to address this gap by supplying a centralized, web app platform where students can seamlessly register, be authenticated, and place water orders from multiple suppliers depending on their availability.

This opportunity enhances convenience, improves service delivery efficiency, supports POPI Act policy, and creates a direct link between students (customers) and water suppliers (service providers). By integrating an automated pricing model (R7 per 5litres), and allowing two payment options, online and cash-on-delivery the system supports financial inclusivity for all users.

The proposed solution aligns with the university's digital transformation initiatives and promotes sustainable campus operations by digitizing routine transactions (Shelly & Rosenblatt, 2012)

(Abu-Alhaija, 2019)

(Barragán-Landy *et al.*, 2020)

FACTS

1. What is your role in the water ordering process?

31 responses

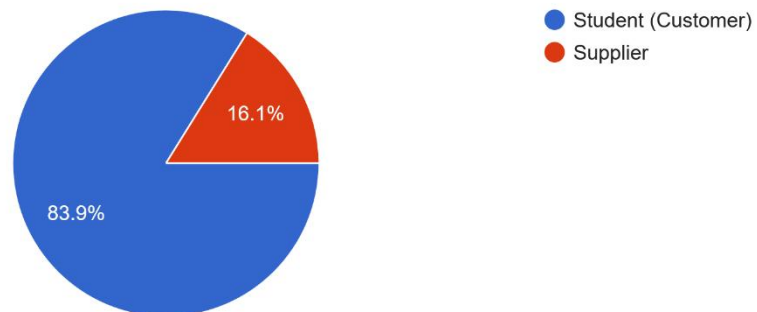


Figure 1:

SUPPLIER

1. Current Ordering and Management a) How do you currently receive orders from students?

5 responses

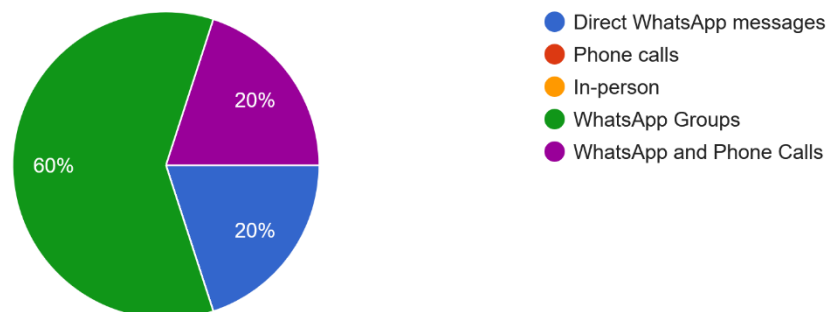


Figure 2:

b) What difficulties do you face when managing orders manually?

5 responses

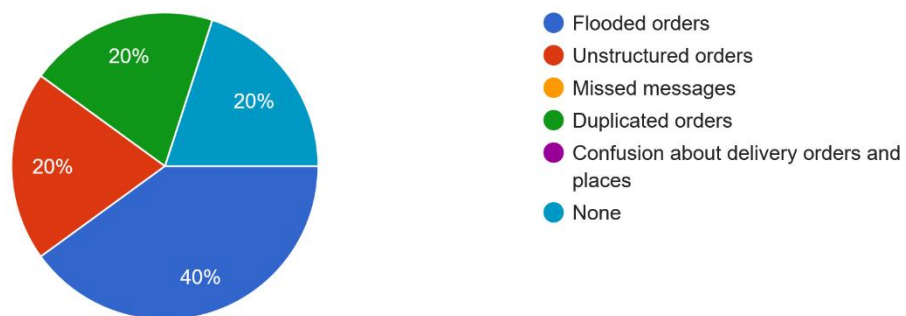


Figure 3:

c) How do you confirm which orders are completed and which are still pending?

5 responses

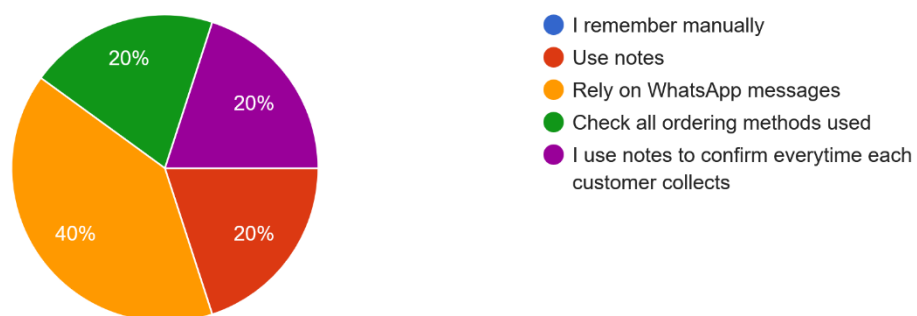


Figure 4:

2. Communication and Confidentiality: a) How do you currently communicate with students about order confirmations or issues?
5 responses

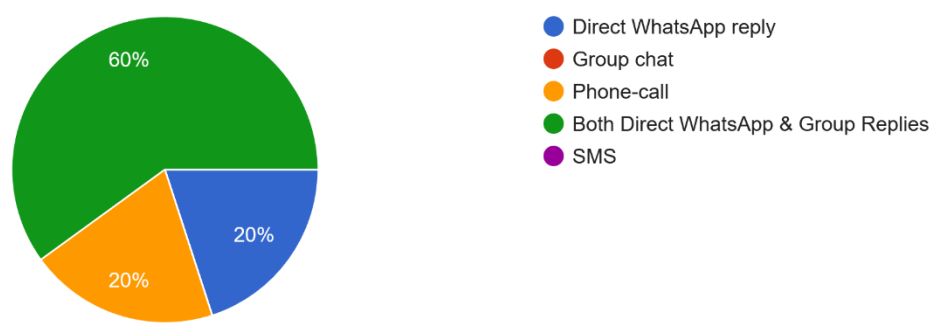


Figure 5:

b) What problems do you face using WhatsApp groups for order communication?
5 responses

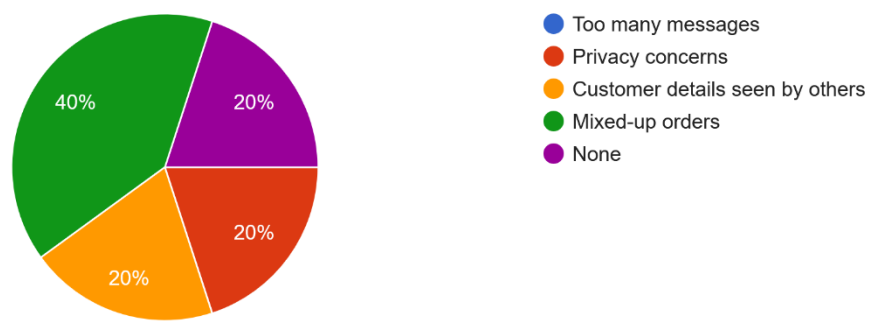


Figure 6:

3. Delivery and Efficiency: a) How do you plan and manage your delivery routes or times?

5 responses

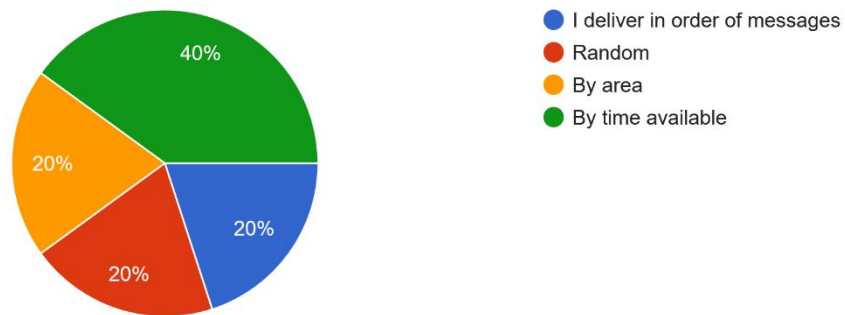


Figure 7:

b) What causes delays or missed deliveries in your current process?

5 responses

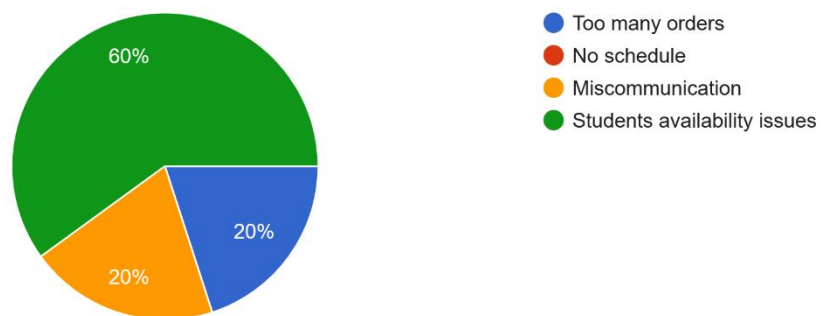


Figure 8:

c) How could a system help you improve delivery scheduling and reduce delays?
5 responses

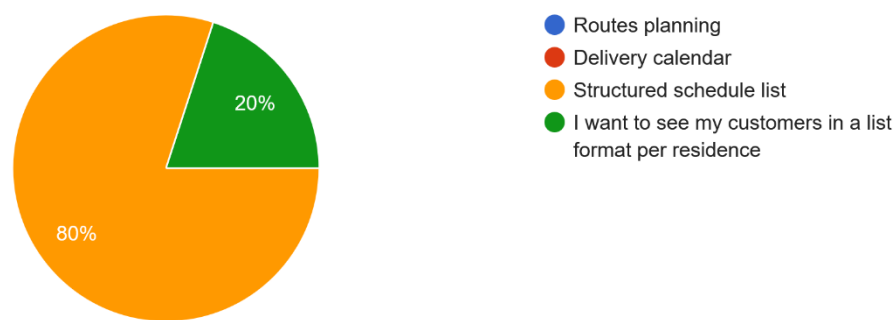


Figure 9:

4. Tracking Sales and Reporting: a) How do you currently track your total sales or income from deliveries?
5 responses

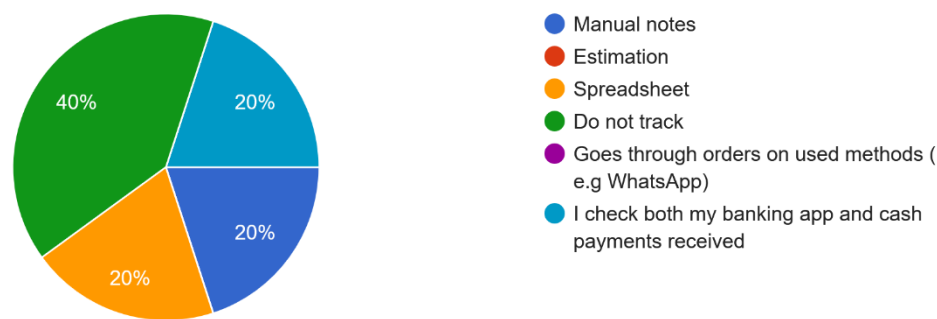


Figure 10:

b) Would an automated dashboard showing daily, weekly, and monthly sales be useful to you? Why?
5 responses

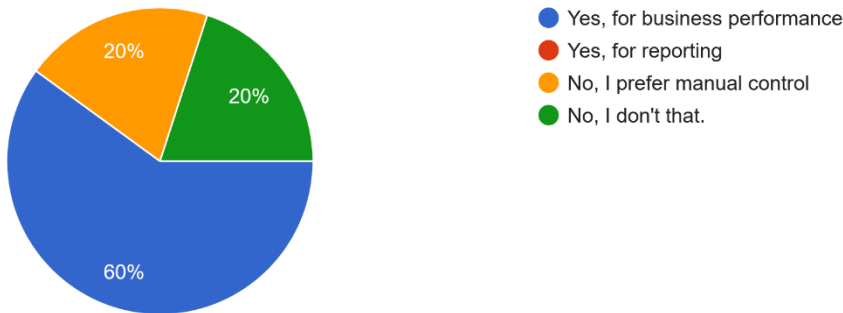


Figure 11:

c) What type of data or analytics would help you run your business better?
5 responses

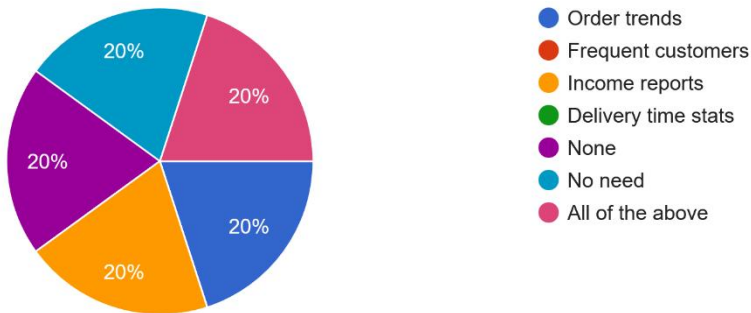


Figure 12:

5. Notifications and System Usability: a) How would you like to receive new order alerts or updates from students?

5 responses

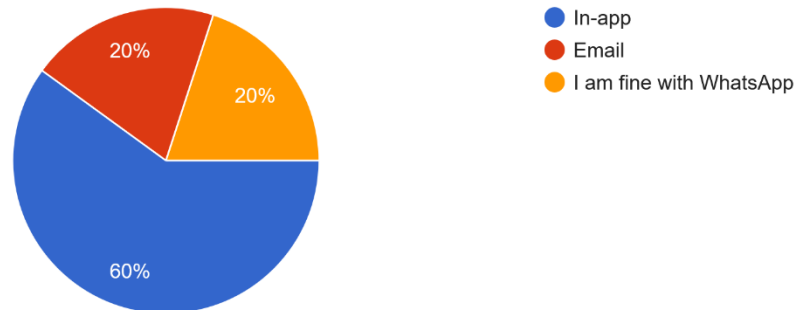


Figure 13:

b) What would make you trust and consistently use a digital ordering system instead of WhatsApp or any other manual method?

5 responses

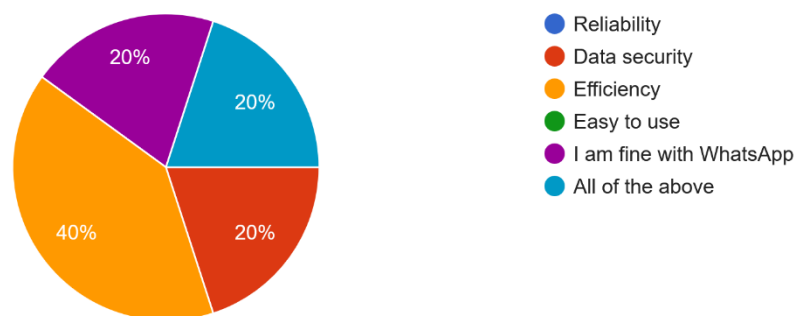


Figure 14:

c) What final recommendations would you make to ensure a centralized ordering system helps both suppliers and students?

4 responses

None. WhatsApp is fine for me

Be accessible and operational at all the time

Ease of use

The system must be user friendly

Figure 15:

FACT FINDING:

STUDENT

1. Current Ordering Process a) How do you currently place an order for purified water?

26 responses

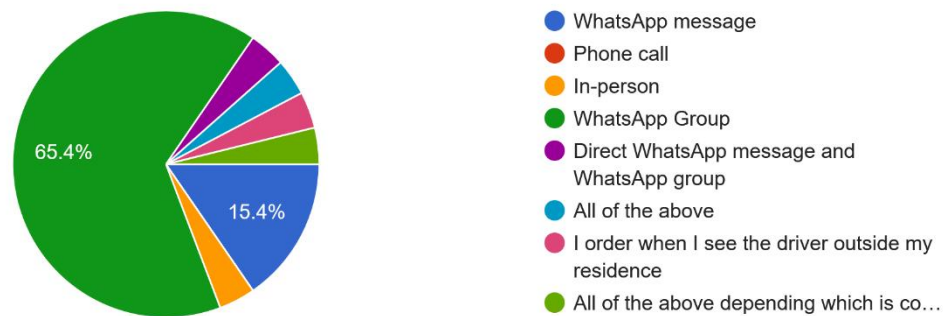


Figure 16:

b) Describe the steps you follow from ordering until receiving your water.

26 responses

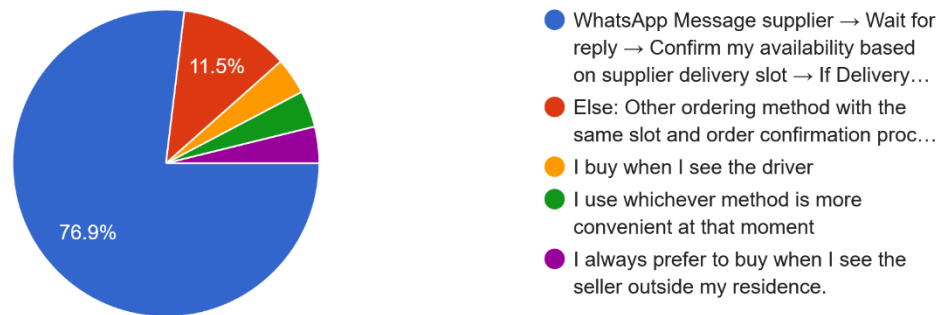


Figure 17:

c) What challenges do you often face when ordering water using WhatsApp or any other method?

26 responses

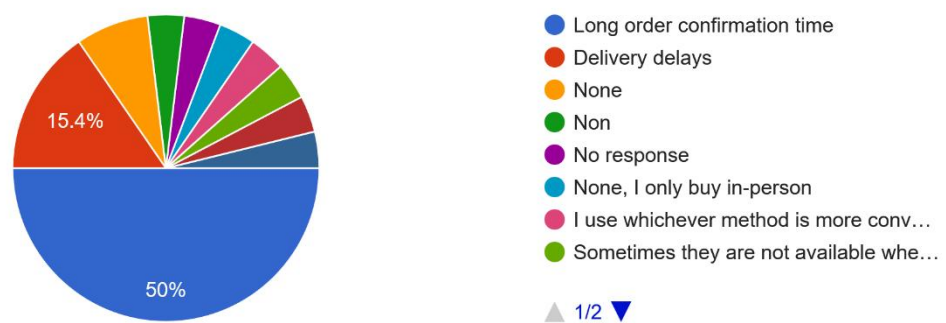


Figure 18:

2. Privacy and Communication: a) How do you feel about your contact number & orders being visible to other students in WhatsApp groups?
26 responses

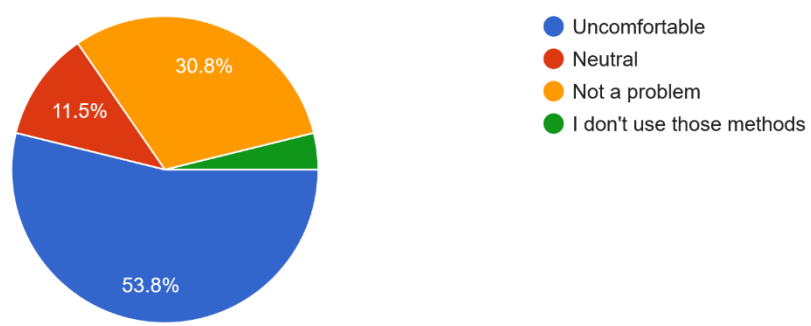


Figure 19:

b) What could be done to make communication between students and suppliers more private and secure?
26 responses

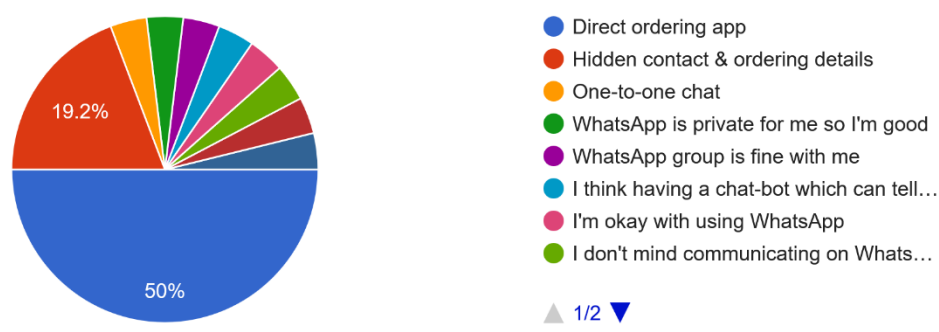
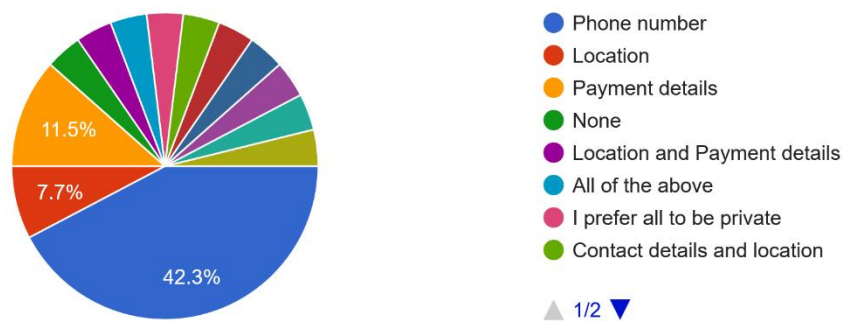


Figure 20:

c) What information do you think should remain confidential when ordering water online?
26 responses



c) What information do you think should remain confidential when ordering water online?

Copy chart

26 responses

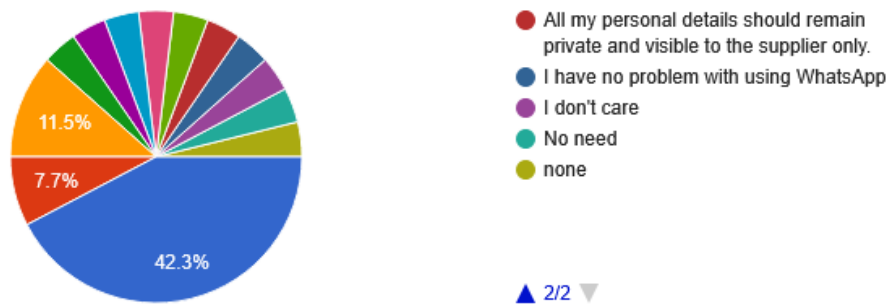


Figure 21:

3. Delivery and Time Management: a) How often do you experience delays or inconsistent delivery times?
26 responses

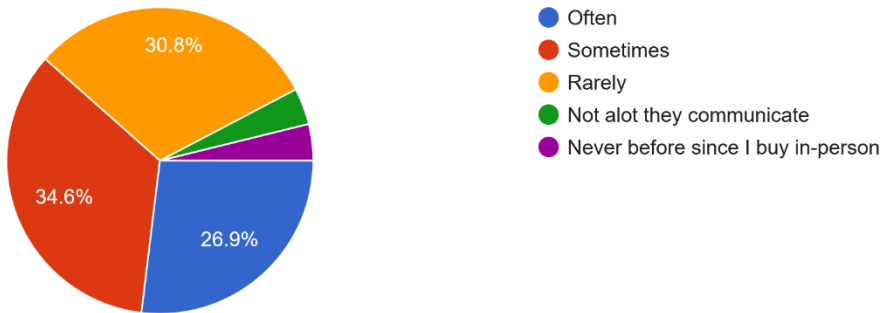


Figure 22:

b) How do late deliveries affect your daily routine, especially when you have classes or other activities?
26 responses

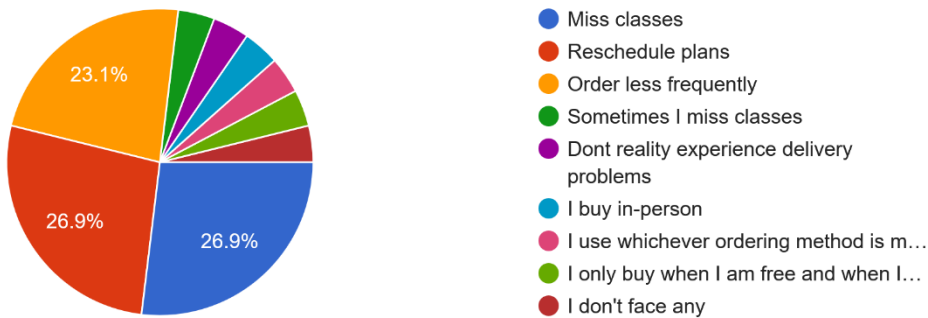


Figure 23:

c) What would help suppliers deliver water more efficiently to students?

26 responses

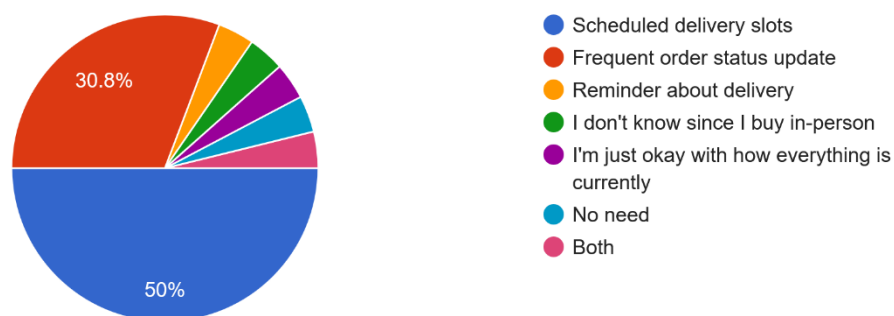


Figure 24:

4. Spending and Tracking: a) How do you currently track how much you spend on water each month?

26 responses

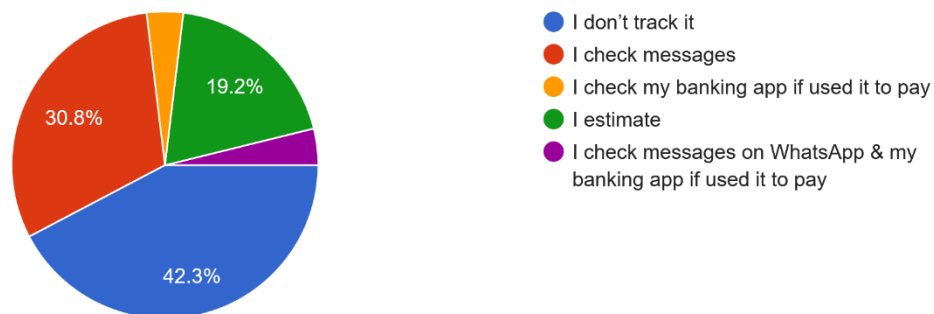


Figure 25:

b) Would a digital system that automatically records your order history and total spending be useful? Why?
26 responses

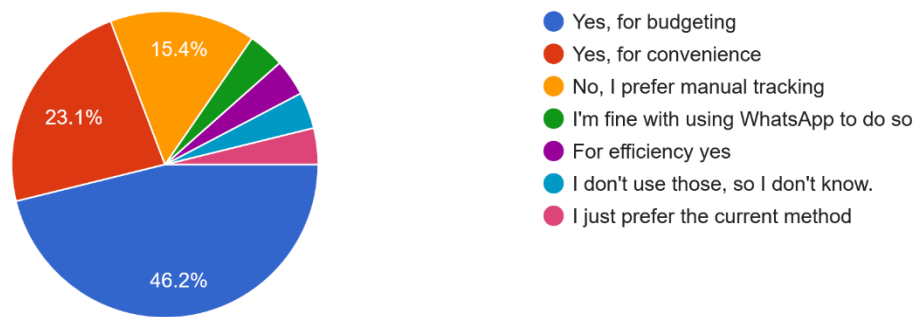


Figure 26:

c) What kind of information would you like to see on your account dashboard?
26 responses

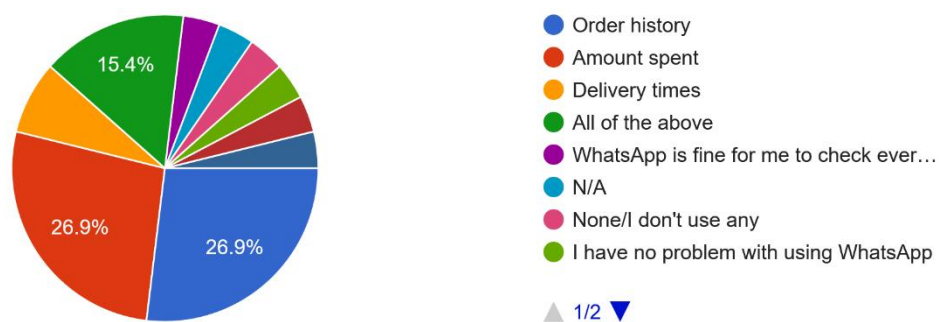


Figure :27

5. Notifications and System Experience: a) How would you like to be notified about your order status (e.g., pending, accepted, on the way, delivered)?

26 responses

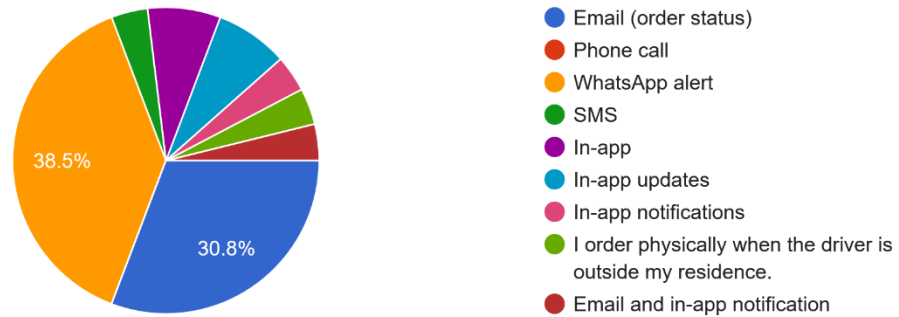


Figure 28:

b) What would make you trust a centralized ordering system to manage your orders better than WhatsApp?

26 responses

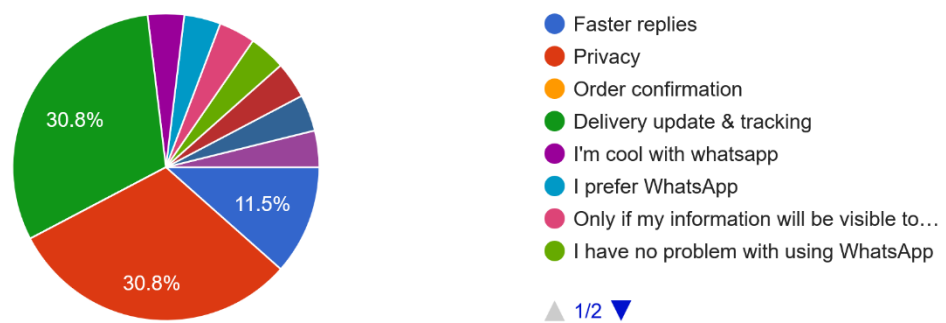


Figure 29:

c) What final suggestions do you have for improving the entire water ordering experience for students?

13 responses

Privacy must be prioritised

Provide a payment method

NO suggestions

No suggestions

Schedule update on advance

Ensure privacy at all costs and improve communication

The supplier must always respond to the orders fast

Need simple and easy order tracking system to improve the current water delivery issues

Privacy matters for me and If I were to use any existing system, I'd prefer my information to be visible to the

Figure 30:

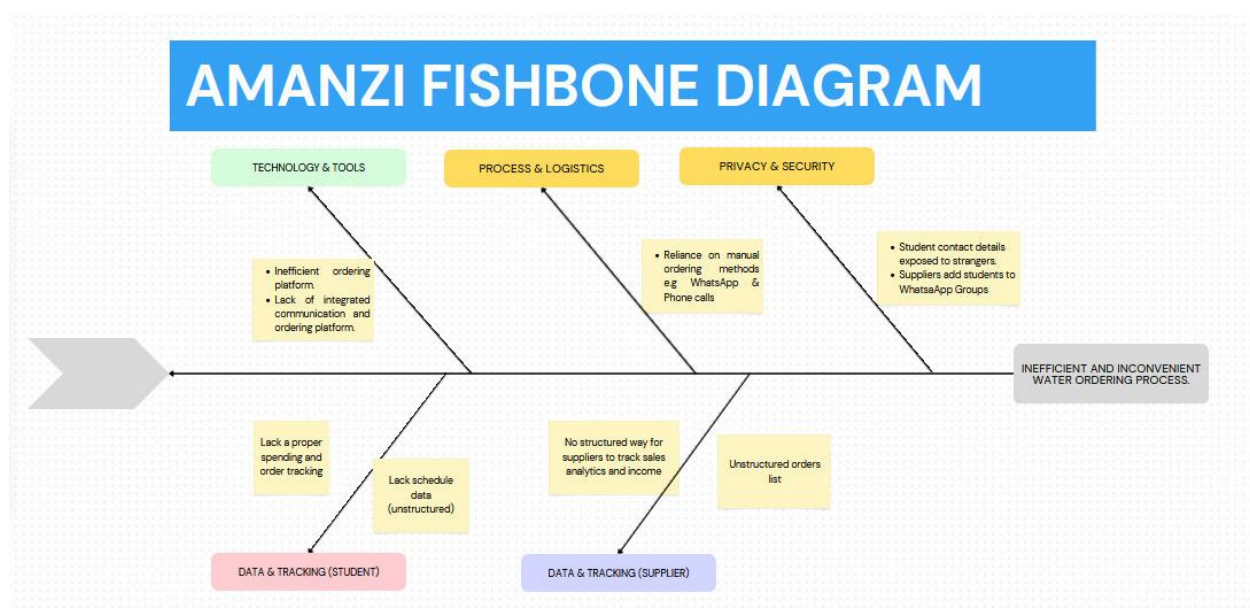


Figure 31: Fishbone diagram

1.2 Step 5 – Evaluate feasibility

The Evaluation of Feasibility assesses whether the proposed Amanzi Water Ordering Information System can be successfully developed, implemented, and maintained within the given constraints of time, technology, budget, and user readiness. This process helps determine if the project is practical, achievable, and beneficial to both students and suppliers at North-West University.

The evaluation focuses on the following key areas:

1.2.1 Operational feasibility

Operationally, Amanzi will streamline the water ordering process by improving efficiency for interaction between students and suppliers. The system operates 24/7, improving accessibility and ensuring accountability in deliveries. Both students and suppliers will benefit from real-time order tracking, reducing miscommunication.

1.2.2 Technical feasibility

The system is a full-stack web app using technologies such as React, CSS, Node.js with Express (backend). A MongoDB Atlas will manage both entities' details and order data. This system also integrates an API for user credential authentication, an email-based "forgot-password" authentication (Mailtrap) and payment gateway API (Stripe) for online transactions.

1.2.3 Economic feasibility

Development costs are minimal since it is built in-house using open-source technologies. Long-term savings include reduced paper usage, lower administrative workload, and increased supplier visibility, which enhances profitability. The return-on-investment (ROI) is expected within one academic year through operational cost reduction (Shelly & Rosenblatt, 2012).

1.2.4 Schedule feasibility

The development process follows a 10-week agile SDLC schedule:

1. Week 1-2: Requirements analysis and design
2. Week 3-5: Frontend and backend development
3. Week 6-7: Database and authentication integration
4. Week 8: Payment and delivery module integration
5. Week 9: Testing and debugging
6. Week 10: Deployment and training

Sampling technique

A purposive sampling method will be used to gather feedback from both on-campus and local water suppliers to ensure system functionality meets user expectations (Barragán-Landy et al., 2020).

Table 1: Work Breakdown Structure - WBS

Task No.	Description	Duration (days)	Predecessor Tasks
1	Requirements gathering and analysis	10	-
2	System design and prototyping	10	1
3	Frontend development	12	2
4	Backend and database integration	14	3
5	Payment gateway and API Integration	7	4
6	Testing and debugging	10	5
7	Deployment and documentation	5	6

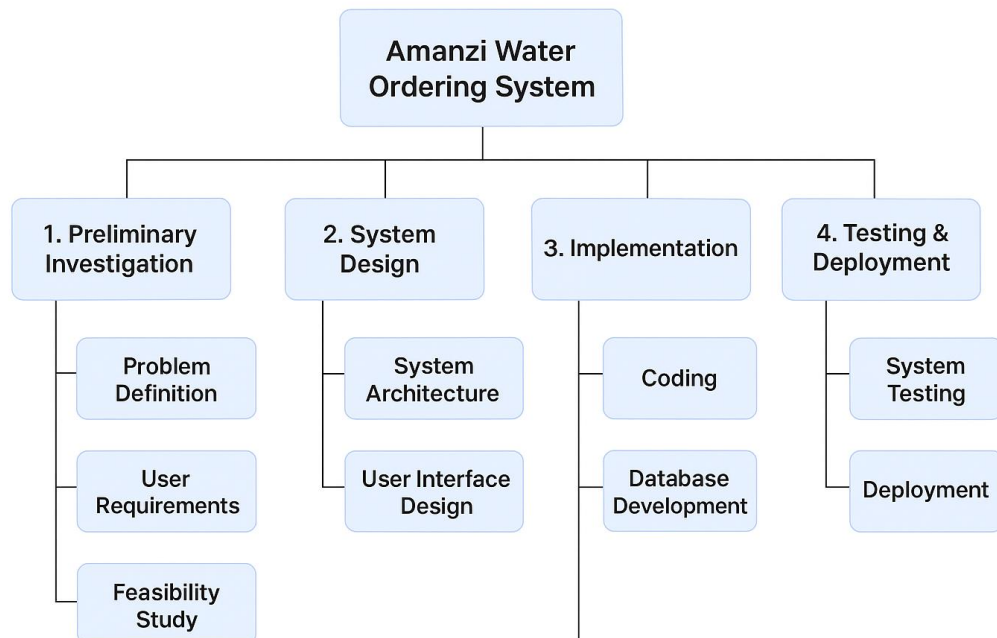


Figure 32: Work Breakdown Structure

PHASE 2 – SYSTEMS ANALYSIS

2 SYSTEM REQUIREMENTS MODELING

2.1 SYSTEM REQUIREMENTS

Table2: STACKHOLDER’S LIST

Name	Role
1. Student/ Customer	Place order in Amazi system
2. Service provider	• Receives Orders • Make deliveries
3. System Developers • BP Mogola • K Makau • TP Sethwape	• Software Developer (Back-end) • Project Manager & System Developer (Front-end) • System Analyst

User Registration and Authentication:

- Both students and suppliers must be able to register accounts, credentials be authenticated, and access “forgot password” authentication.
- Users must choose their role: Student or Supplier.
- Login credentials must be verified against stored data before access is granted.

Student Functions:

- Register.
- Login upon registration.
- Place orders specifying water quantity (5L minimum).
- View price automatically calculated at R7 per 5litres.
- Choose payment method: Online Payment or Cash-on-Delivery.
- Download or view order invoice (pdf).
- Track order status: Pending, Accepted, Completed.
- Access order summaries.

Supplier Functions:

- Register.
- Login to supplier interface.
- View and manage orders (Accept/Reject/Complete).
- Give a reason in case of rejection.
- Monitor revenue and total water sold.
- View analytics dashboard reflecting total sales and order categories.
- Refresh dashboard to view update data.

Non-Functional Requirements:

- **Performance:** The web app handles concurrent access by 100+ users without delays.
- **Security:** All passwords must be hashed: email authentication (forgot password) and payment data are secure SSL/TLS connections.
- **Usability:** System is intuitive, mobile-responsive, and accessible to all students.
- **Scalability:** System should accommodate future expansion to additional NWU Mahikeng off-campus residences.
- **Availability:** System uptime should exceed 99% during high demand periods.

2.1.1 Output

Expected system outputs include:

- Student dashboards displaying order history, status, and total purchases.
- Supplier analytics dashboard with total revenue, water sold, and order distribution.
- Downloadable PDF invoices for every transaction.
- System-generated notifications for accepted, rejected, delivery status, or completed orders.

2.1.2 Security

Security controls include:

- Email verification “forgot password”.
- Password encryption.
- Role-based access control (Student/Supplier)
- HTTPS for secure communication.
- Database access restricted by user role (Only DB Admins).

2.2 DATA AND PROCESS MODELING

DATA FLOW DIAGRAM:

Student

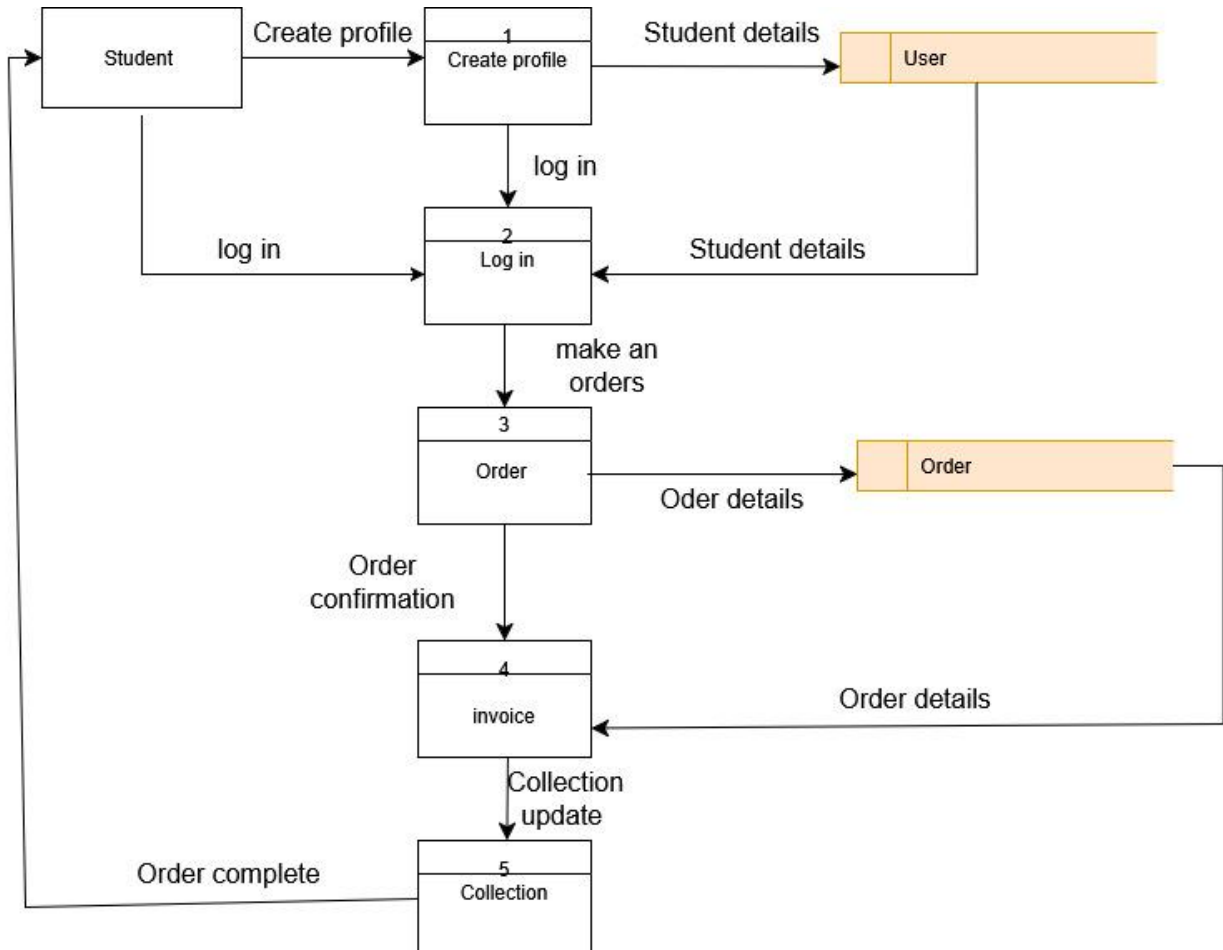


Figure 33: DFD Student/ customer

DATA FLOW: Service Provider

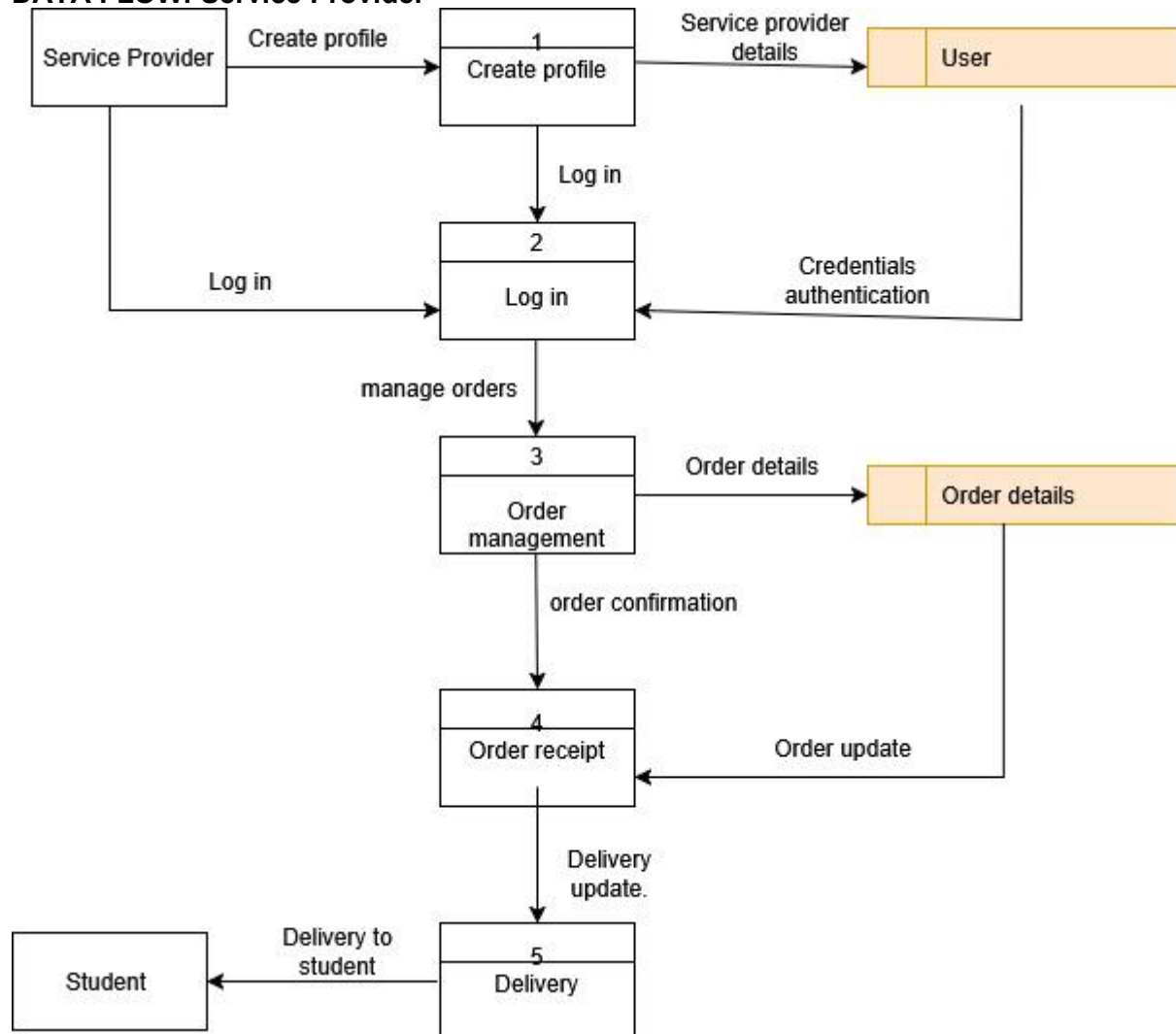


Figure 34: DFD Service provider

USE CASE DIAGRAM:

Amanzi Use Case Diagram

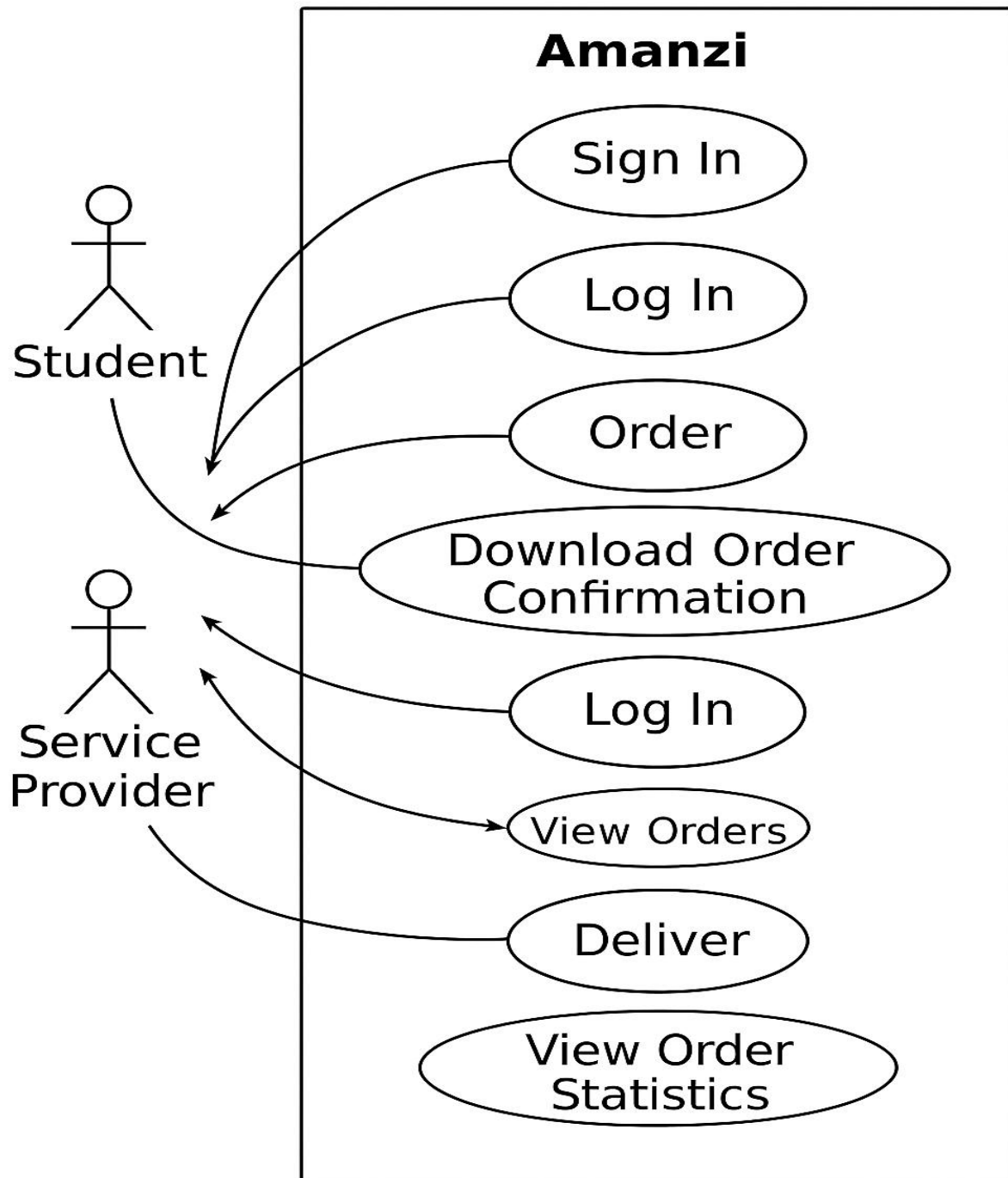


Figure 35: Use Case Diagram

USE CASE DESCRIPTION FOR CUSTOMERS

Table 3: New customers sign up case Description

Use case ID:	01	Use case Name:	Student creates a new profile
Priority	01		
Actor	Customer/ student		
Use Case Summary:	It allows the user (Students) to register themselves with the system by providing the information like name, valid email.		
Pre-conditions	The email should be valid.		
Normal Follow of Paths		Alternative Path	
1. The user clicks on sign up button.			
2. The user gives the information like name, email, and password.			
3. A verification mail is sent to the provided email.		If user does not click on sign up then verification email will not be sent.	
Exception:			
1. The system do not allow the user to add email which is already registered with in the system.			
Post-conditions:			
Step no #	Description		
1.	The user (Student) will be successfully registered.		

Table 4: Customers Log in use case for customers

Use case ID:	02	Use case Name:	Log in
Priority	02		
Actor	Customer/ Student		
Use Case Summary:	It allows the users (Customers) to login themselves into the system using the email and password which they provide at the time of registration		
Pre-conditions	User should be registered with the system.		
Normal Follow of Paths		Alternative Path	
1. The user clicks on log in button.			
2. The user provides the email and password.			
Exception:			
1. If the email or password is not valid then the user will not be able to be logged in with the system			
Post-conditions:			
Step no #	Description		
1.	The user (Customer, Student) will be successfully logged in.		

Table 5: Order case description

Use case ID:	03	Use case Name:	Order
Priority	01		
Actor	Customer/ Student		
Use Case Summary:	It allows the users (Customers) to order water into the system		
Pre-conditions	Owner should be registered and logged in with the system before ordering In the system.		
Normal Follow of Paths		Alternative Path	
1. The user clicks on order button.			
2. The customer provides the information like name, email, litres, location(residence) and selects schedules			
4. After entering all the information, the customer/ student will press the submit button			
5. After pressing the submit button an automated mail is sent to the provided email with a unique reference number.			
Exception:			
The order will be sent to the customer for the delivery			
Post-conditions:			
Step no #	Description		
1.	The user (Customer/ Student) order will be successfully placed		

Table 6: View order case description

Use case ID:	04	Use case Name:	View Order
Priority	01		
Actor	Customer/Student		
Use Case Summary:	It allows the user (Customer/ student) to check the status of their order		
Pre-conditions	The order should be placed into the system.		
Normal Follow of Paths		Alternative Path	
1. The user will click on the view status button.			
2. The users will enter the reference number of their order.			
3. The user will press the enter button.		The user will be able to see the status if their order is accepted or declined by the service provider.	
Exception:			
If the reference number is not valid, the system will not show any status.			
Post-conditions:			
Step no #	Description		
1.	The status of the item will be displayed on the screen.		

Table 7: View statistics

Use case ID:	05	Use case Name:	View statistics
Priority	04		
Actor	Customer, Student		
Use Case Summary:	The Customer, Student can view the statistics of all the number of orders placed and received		
Pre-conditions	Customer student must be logged in to view the statistics		
Normal Follow of Paths		Alternative Path	
1. The customer/ Student logs into the system with valid inputs		If the input information provided is not correct, the admin is to be able to view statistics	
2. The customer/ Student can view the statistics on summary			
Exception:			
Post-conditions:			
Step no #	Description		
1.	The customer/ Student will successfully view the statistics.		

CASE DESCRIPTIONS FR THE SERVICE PROVIDERS

Table 8: New Service Provider sign up case Description

Use case ID:	05	Use case Name:	Student creates a new profile
Priority	01		
Actor	Service Order		
Use Case Summary:	It allows the user service provider to register themselves with the system by providing the information like name, valid email.		
Pre-conditions	The email should be valid.		
Normal Follow of Paths		Alternative Path	
1. The user clicks on sign up button.			
2. The user gives the information like name, email, and password.			
3. A verification mail is sent to the provided email.		If user does not click on sign up, then verification email will not be sent.	
Exception:			
1. The system do not allow the user to add email which is already registered with in the system.			
Post-conditions:			
Step no #	Description		
1.	The user (Service Provider) will be successfully registered.		

Table 9: Log in case description for service providers

Use case ID:	06	Use case Name:	Log in
Priority	02		
Actor	Service Provider		
Use Case Summary:	It allows the users (Service Provider) to login themselves into the system using the email and password which they provide at the time of registration		
Pre-conditions	User should be registered with the system.		
Normal Follow of Paths		Alternative Path	
1. The user clicks on log in button.			
2. The user provides the email and password.			
Exception:			
1. If the email or password is not valid then the user will not be able to be logged in with the system			
Post-conditions:			
Step no #	Description		
1.	The user (Service Provider) will be successfully logged in.		

Table 10: View orders case description

Use case ID:	07	Use case Name:	View Orders
Priority	01		
Actor	Service Provider		
Use Case Summary:	It allows the user (Service Provider) to check the status of their order		
Pre-conditions	The order should be placed into the system.		
Normal Follow of Paths		Alternative Path	
1. The user will click on the view status button.			
2. The users will accept or decline the orders.			
3. The user will press the accept/decline button		The user(customer/Student) will be able to see the status if their order is accepted or declined by the service provider.	
Exception:			
If the reference number is not valid, the system will not show any status.			

Post-conditions:	
Step no #	Description
1.	The status of the item will be displayed on the screen.

Table 11: View Historical orders

Use case ID:	08	Use case Name:	View History
Priority	01		
Actor	Service Provider		
Use Case Summary:	Service Provider will be able to access the number of accepted orders to deliver from the placed orders		
Pre-conditions	Website must be opened to access the list of the orders		
Normal Follow of Paths		Alternative Path	
1. The user clicks on the recent orders		If the user(Service provider) is not logged in, then he/she will be redirected to login page	
2. The user see the pending orders to deliver.			
Exception:			
User will be able to see the ordered customers			
Post-conditions:			
Step no #	Description		
1.	The user will successfully see his items history.		

Table 12: View statistics

Use case ID:	09	Use case Name:	View statistics
Priority	04		
Actor	Service provider		
Use Case Summary:	The service provider is able to view the statistics of all the received, deliveries and rejected orders they had		
Pre-conditions	Service provider must be logged-in to view the statistics		
Normal Follow of Paths		Alternative Path	

2. The service provider logs into the system with valid inputs	If the input information provided is not correct, the admin is to be able to view statistics
2. The service provider can view the statistics on Orders summary	
Exception:	
Post-conditions:	
Step no #	Description
1.	The service provider will successfully view the statistics.

PHASE 3 - SYSTEMS DESIGN

3 SYSTEM DESIGN SPECIFICATION

3.1 Overview

This design specification translates the functional and non-functional requirements from Phase 2 into detailed system components, database designs, user interface designs, and architecture diagrams. The design follows the structured SDLC guidance of Shelly & Rosenblatt (2012) and is tailored for a full-stack web application used by Students and Water Suppliers at NWU Mahikeng. The system supports two payment methods: Online Payment and Cash-on-Delivery.

Primary goals

- Usable, responsive web UI for Students and Suppliers.
- Secure registration.
- Secure forgot password authentication with email.
- Reliable order lifecycle (Pending → Accepted/Rejected → Completed)
- Supplier analytics dashboard (revenue, water sold)
- Invoice generation (PDF) per order
- Scalable architecture for future campus expansion (Shelly & Rosenblatt, 2012).

Tech stack (recommended)

- Frontend: React + HTML5, CSS3, responsive design (Tailwind)
- Backend: Node.js + Express
- Database: MongoDB Atlas
- Authentication: email for password update.
- PDF generation: server-side library
- Hosting: Netlify

3.2. Software Architecture

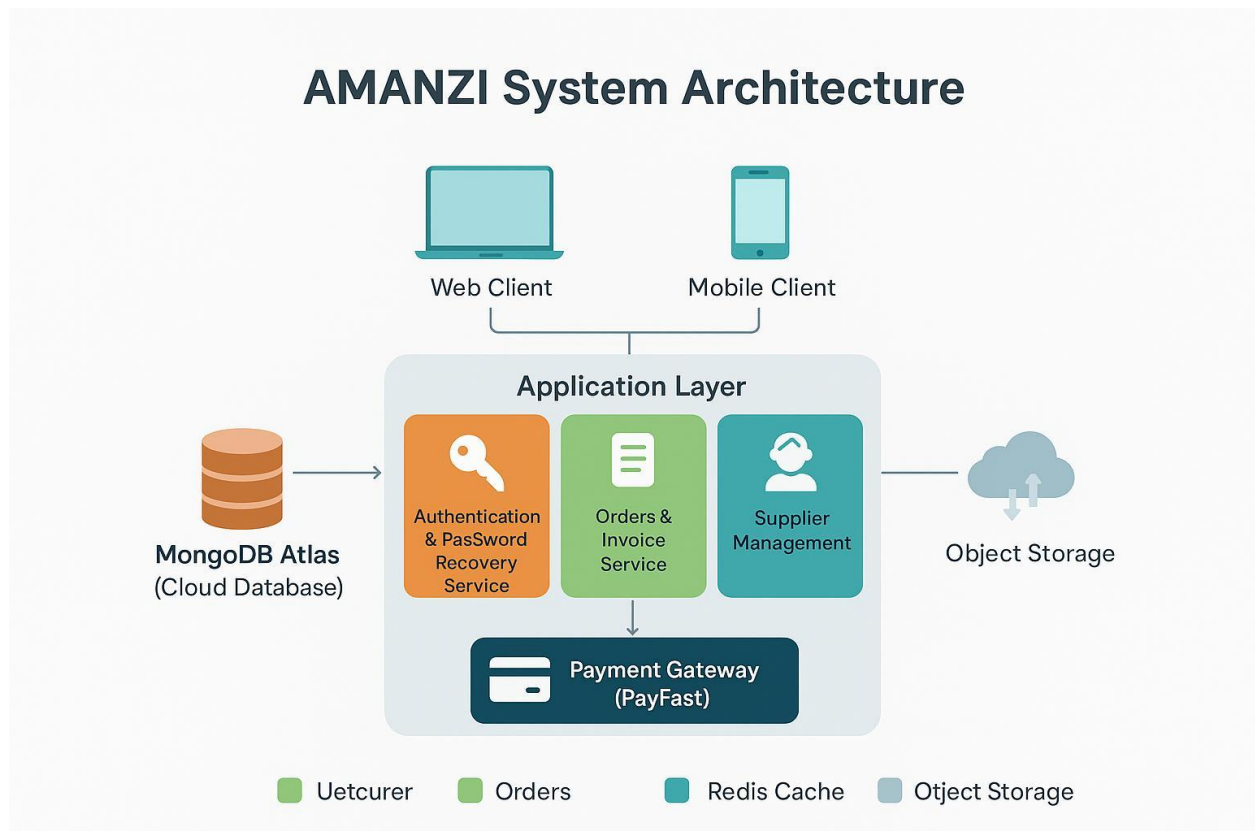


Figure 36: Software Architecture

3.3 User Interface Design

Design principles applied:

- Mobile-first and responsive; clear affordances for role choice (Student vs Supplier).
- Left-hand vertical navigation for role-specific features
- Minimal steps to place an order; price updates live when quantity changes.
- Prominent call-to-action: Place Order and invoice generation.

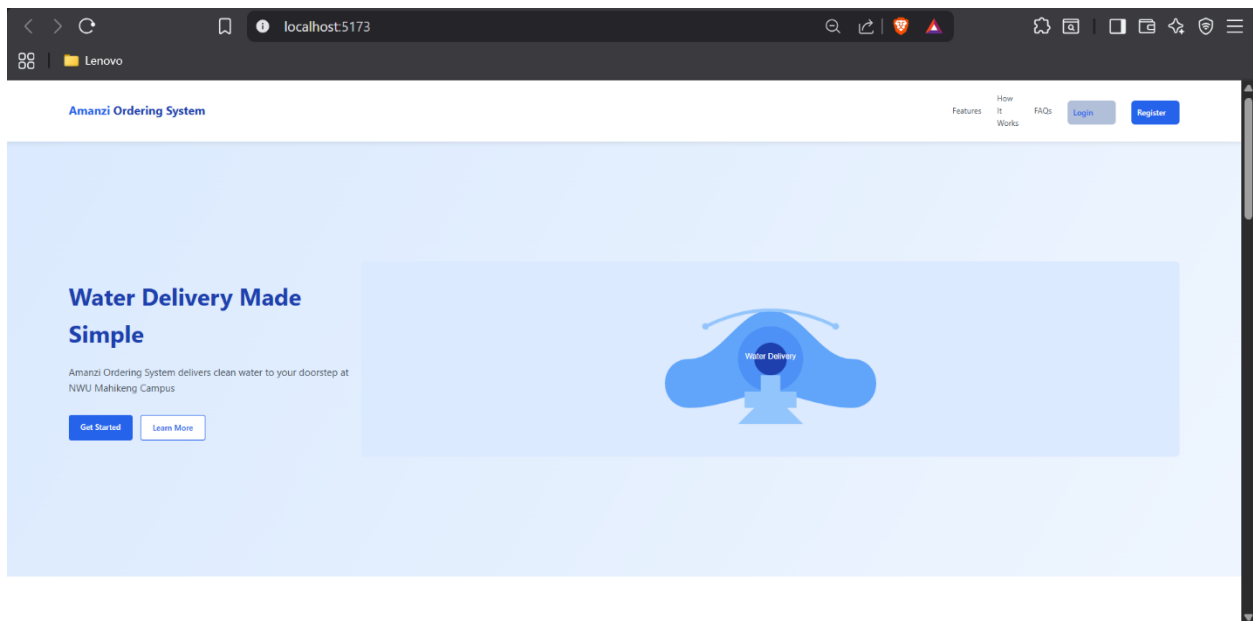


Figure 37: Welcome Home Page

A screenshot of a web browser displaying the 'localhost:5173/register' page. The page features a central white registration form titled 'Create Your Account'. The form includes the following fields and options: a sub-header 'Join Amanzi for water delivery', a 'Full Name' input field, an 'Email' input field, a 'Password' input field, a 'Confirm Password' input field, a section for 'I am a:' with 'Student' and 'Water Provider' radio buttons, a 'Residence' dropdown menu with the text 'Select your residence', and a blue 'Register' button. At the bottom of the form, there is a link that says 'Already have an account? Login'.

Figure 38: Register form

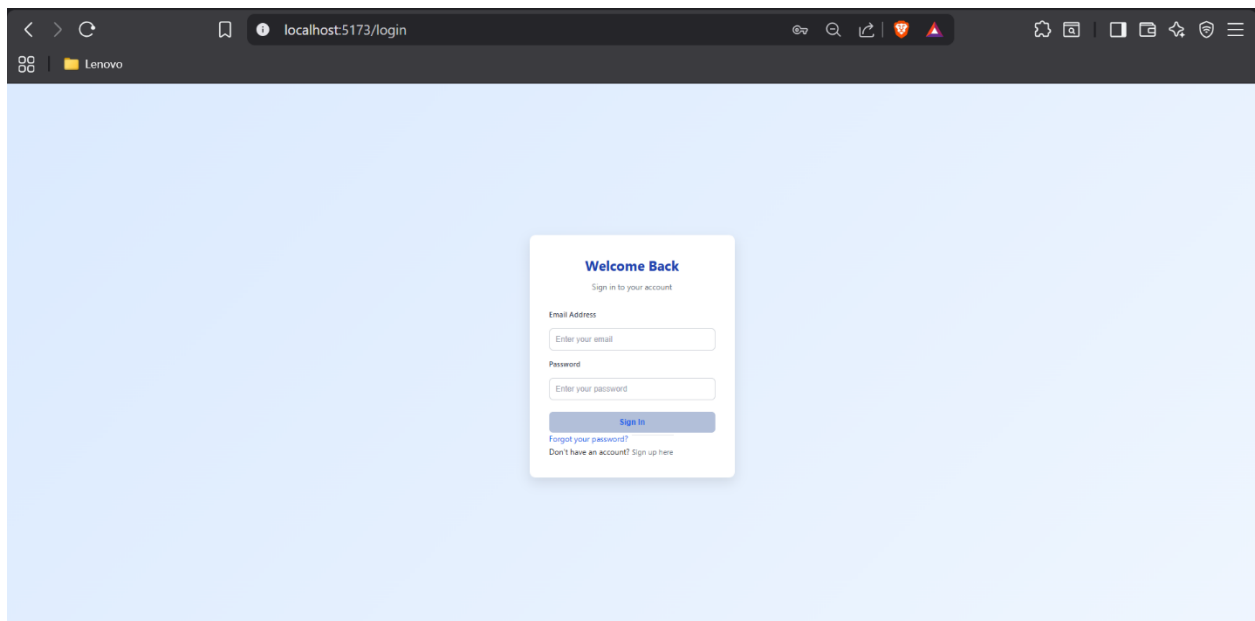


Figure 39: Login form

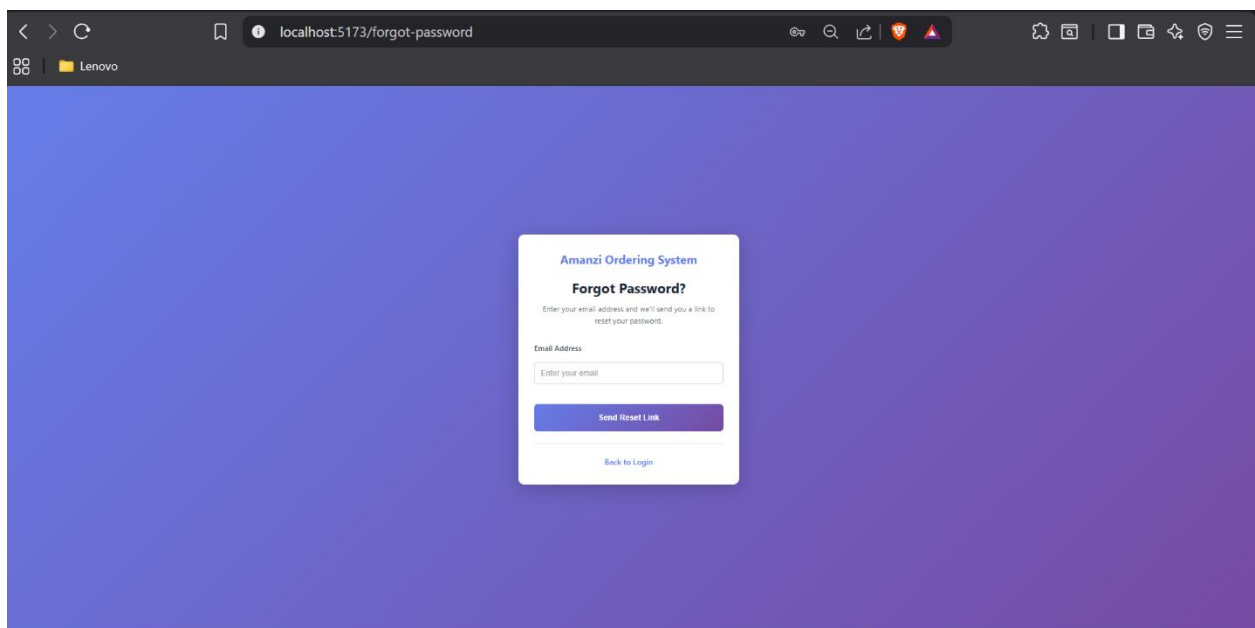


Figure 40: Forgot Password

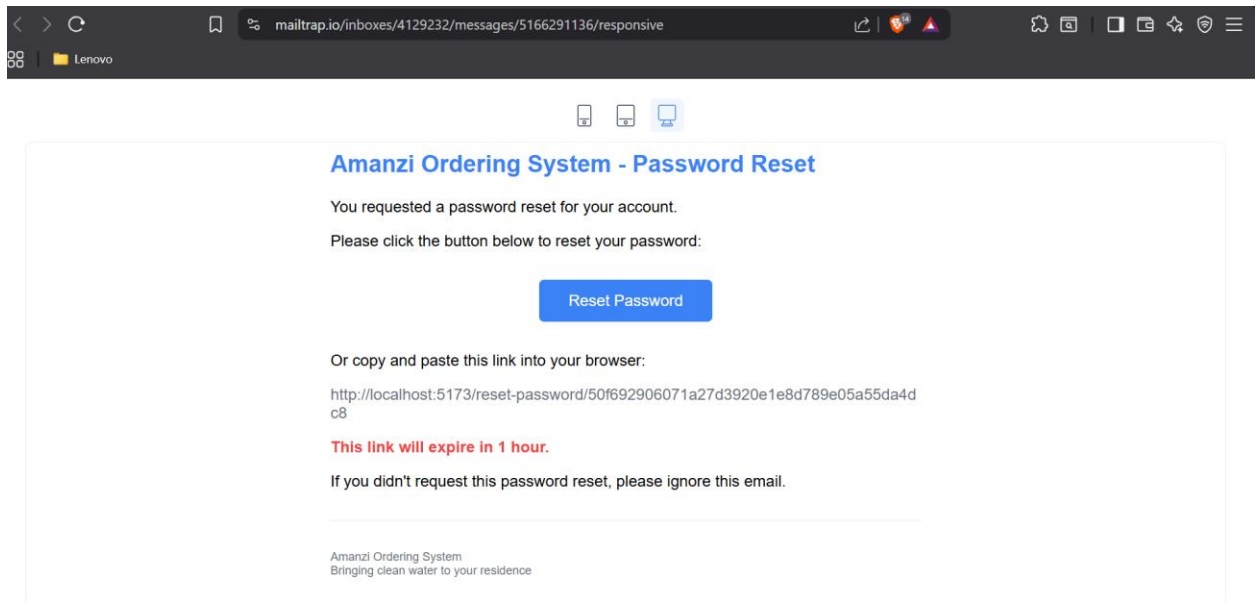


Figure 41: Reset Password email

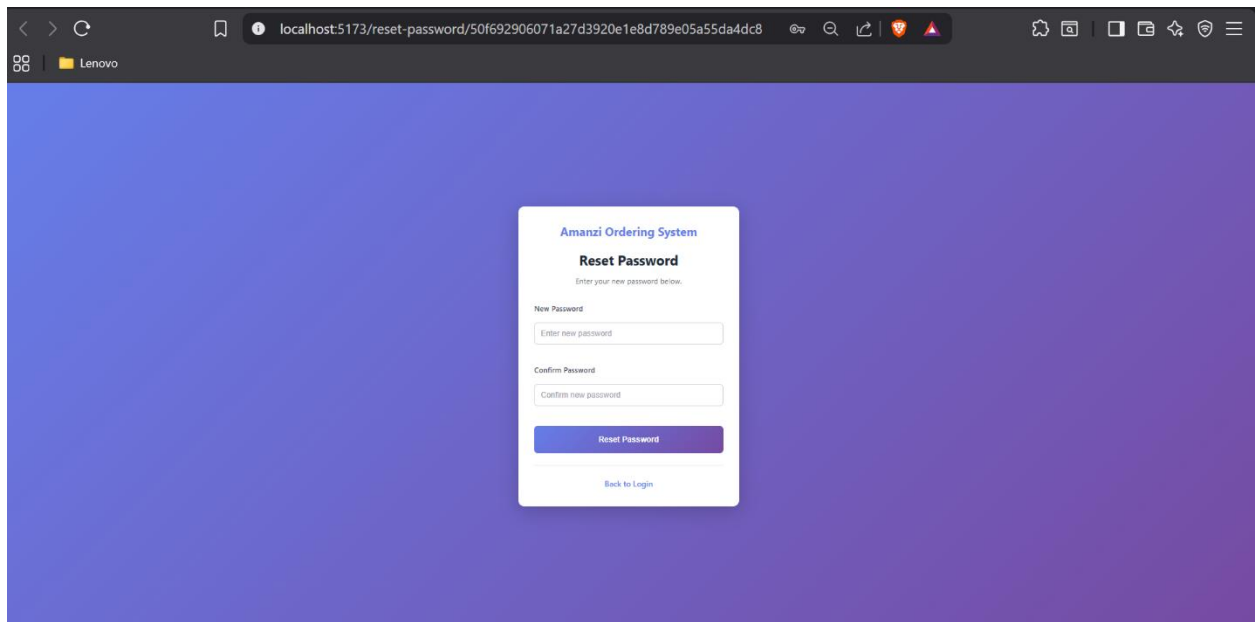


Figure 42: Reset Password

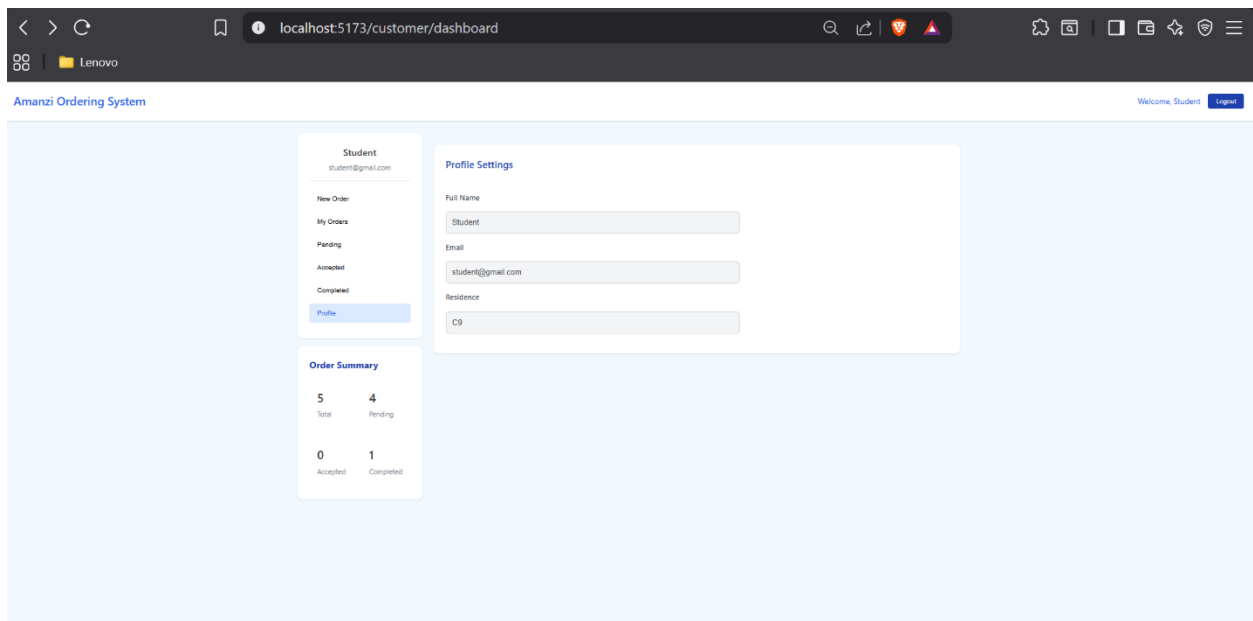


Figure 43: Customer Profile

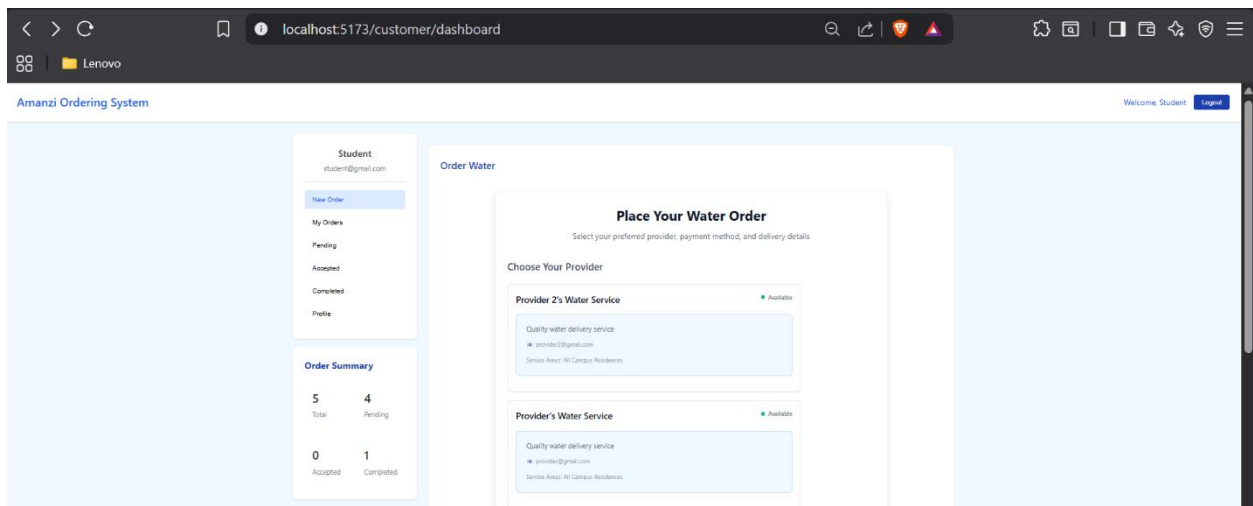


Figure 44: Customer Dashboard

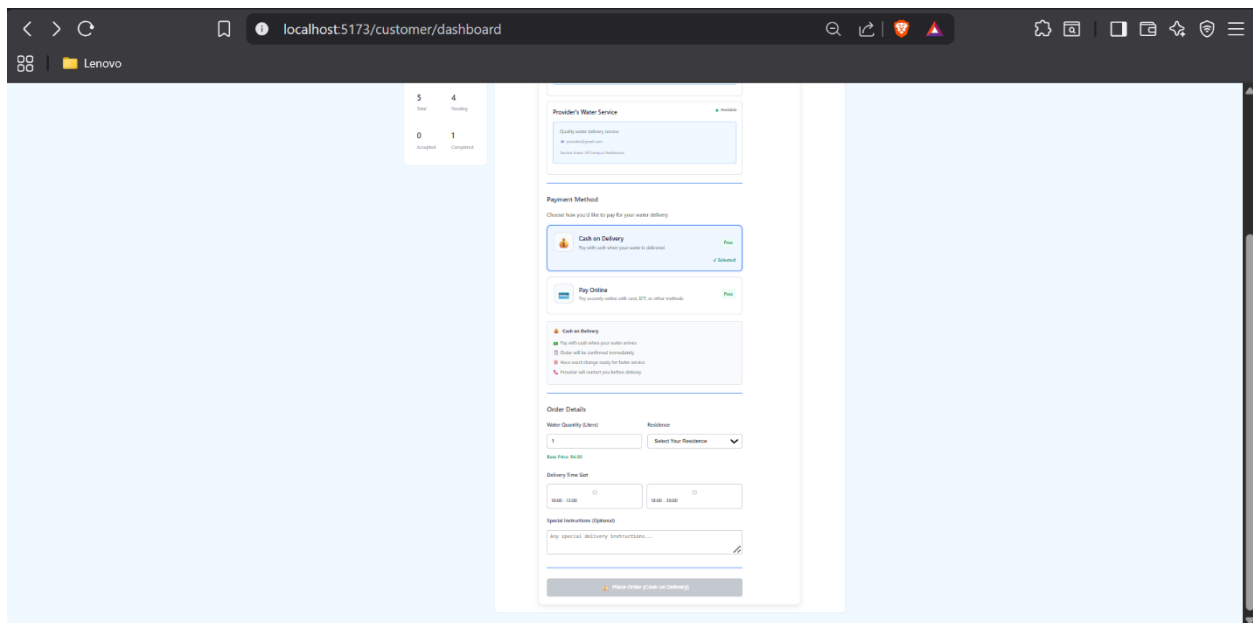


Figure 45: Customer Dashboard Cont...

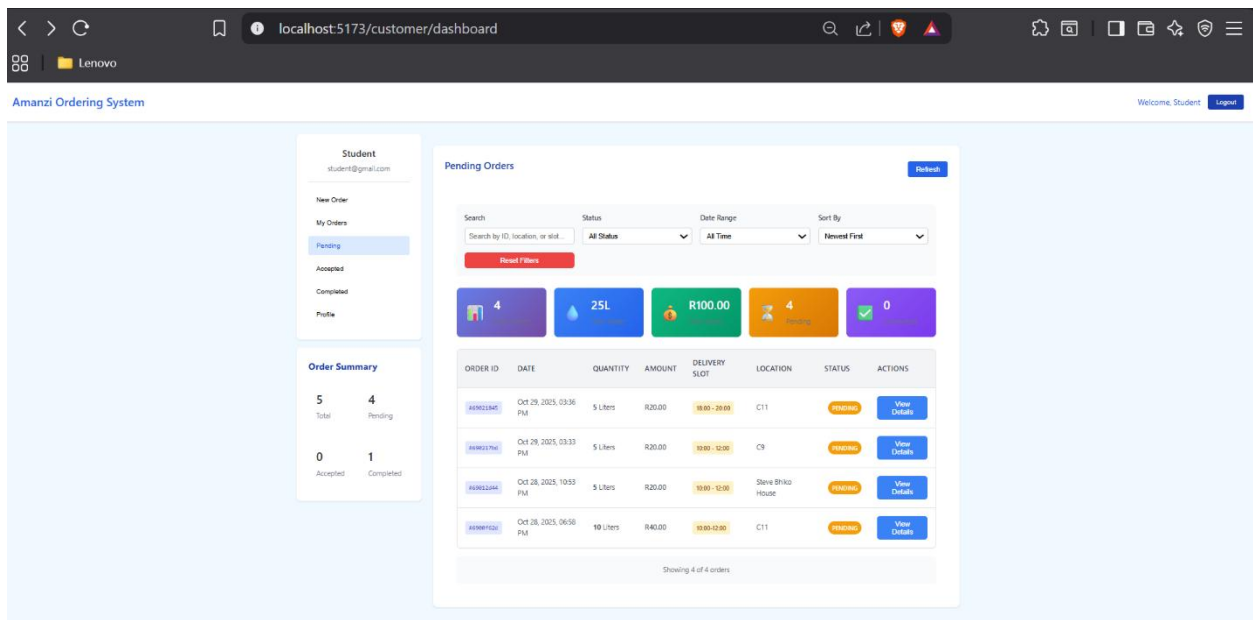


Figure 46: Customer Pending Orders

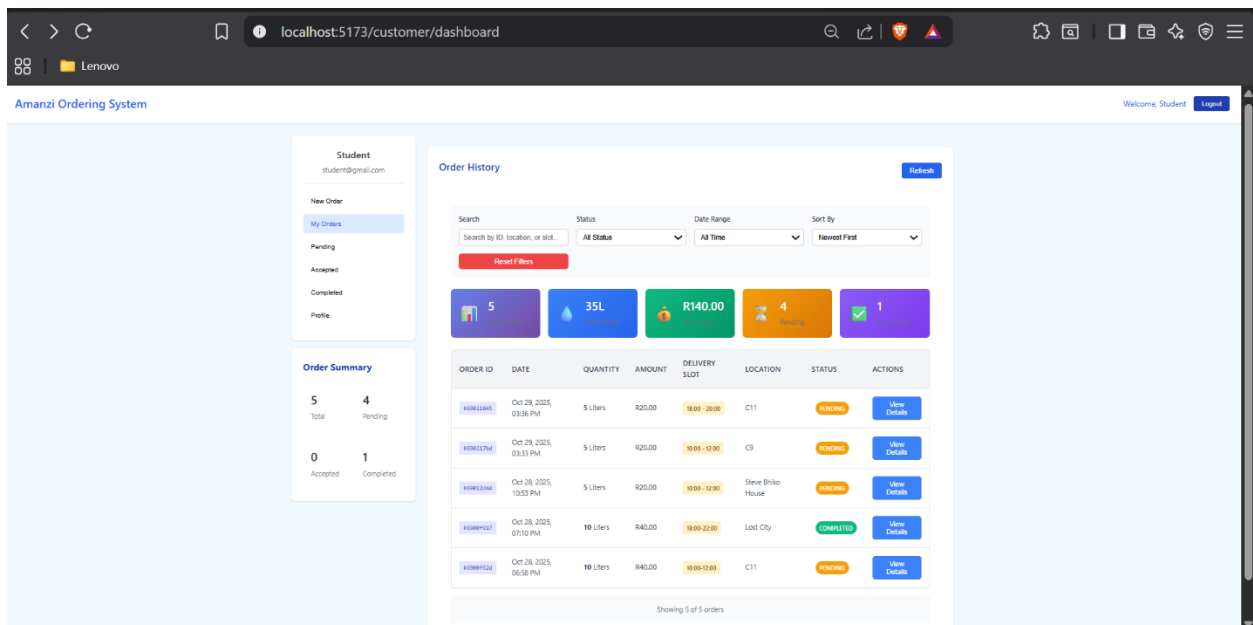


Figure 47: Customer Order History

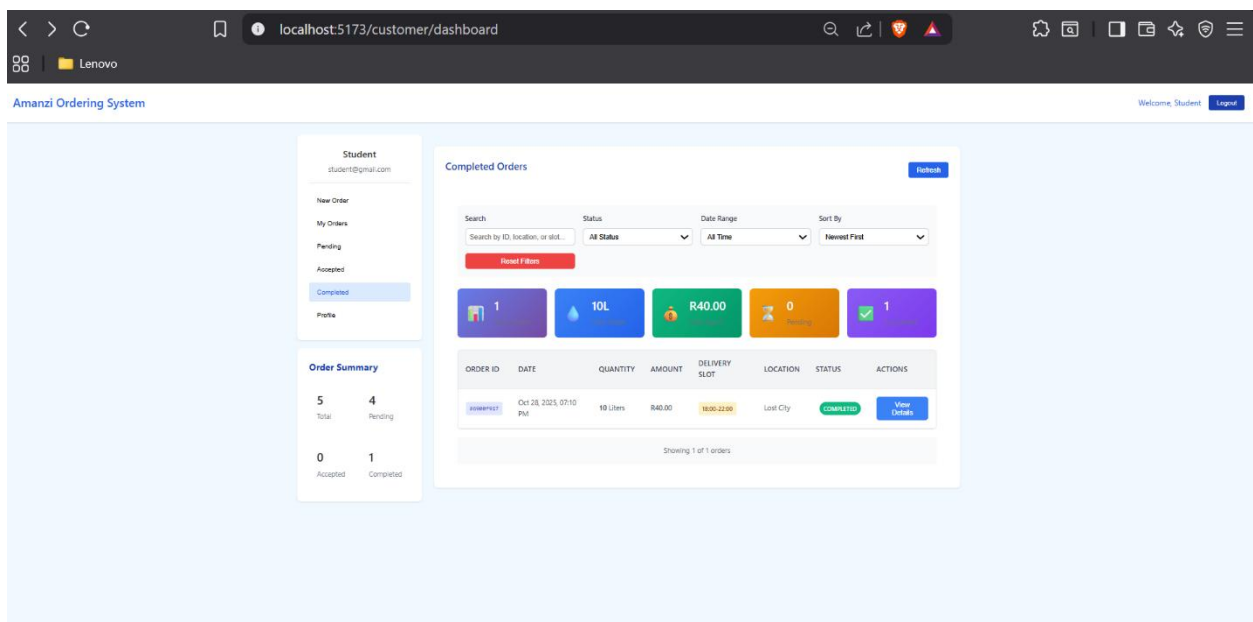


Figure 48: Customer Completed Orders

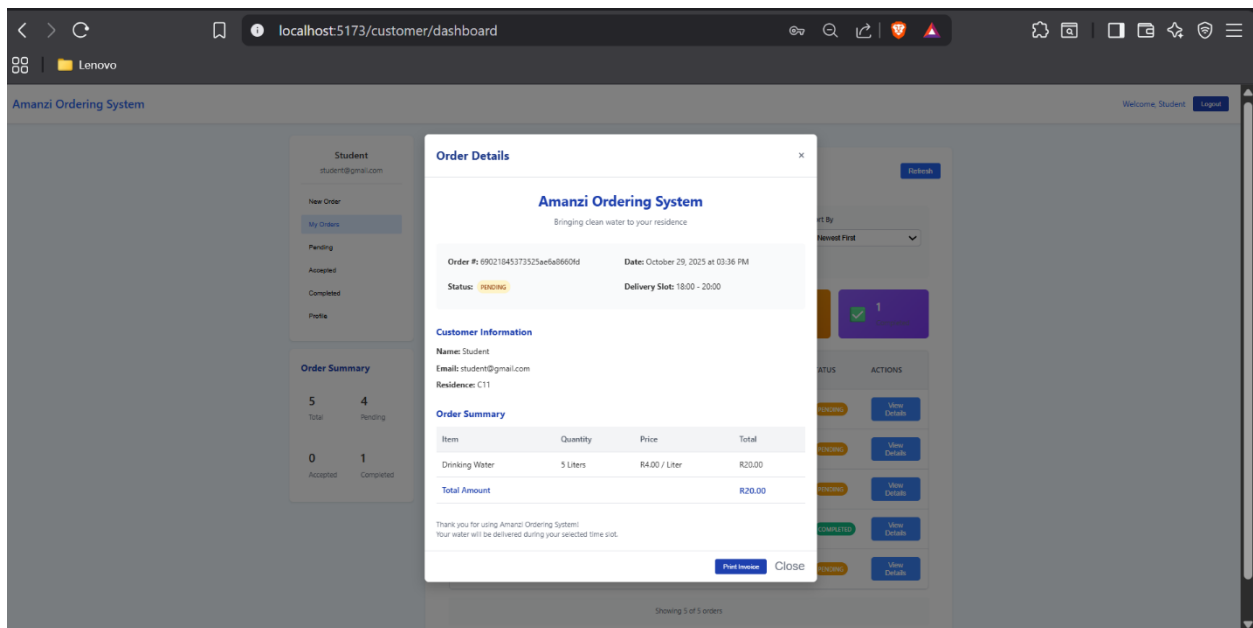


Figure 49: Customer Order Details/Invoice

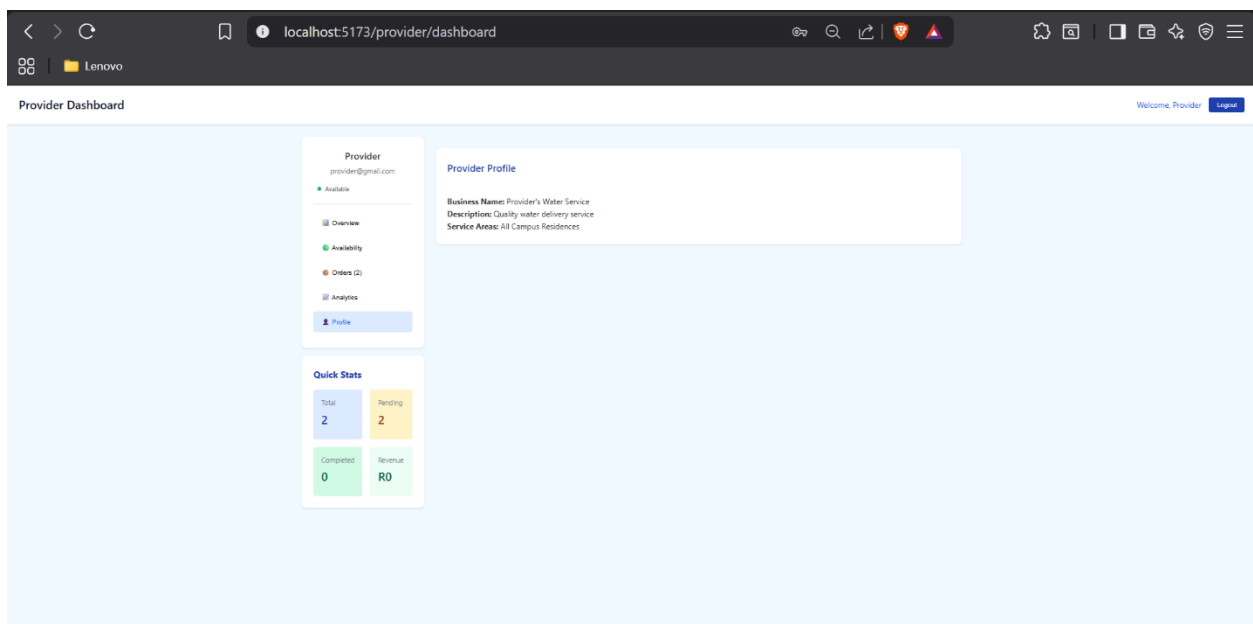


Figure 50: Provider Profile

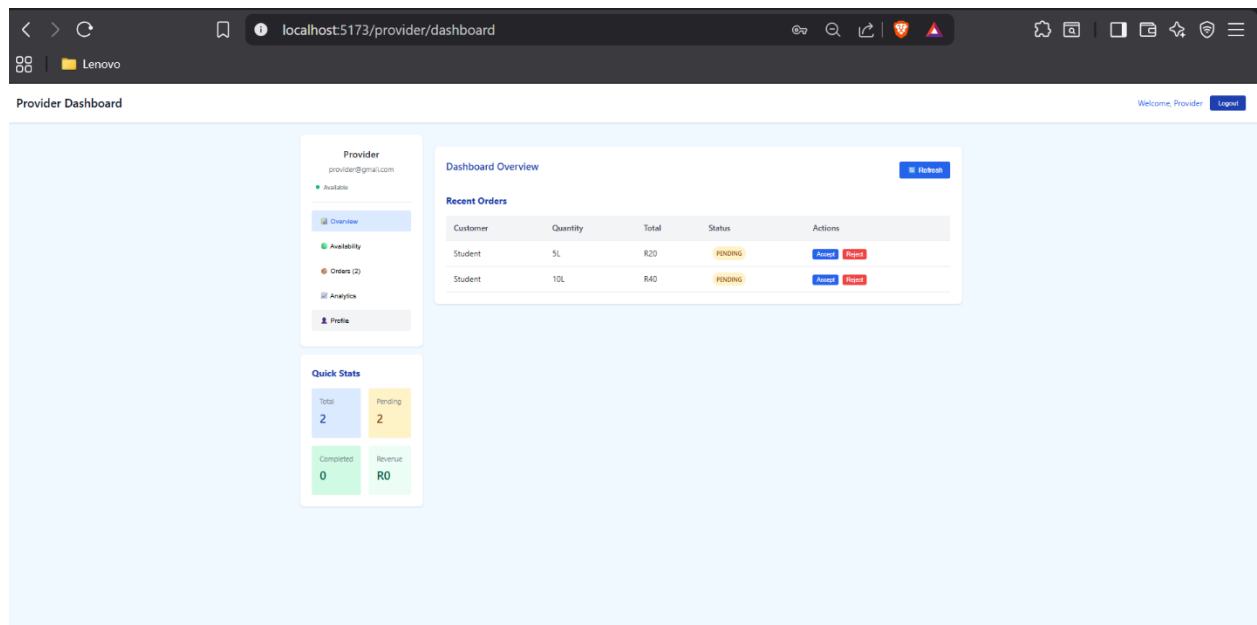


Figure 51: Provider Dashboard

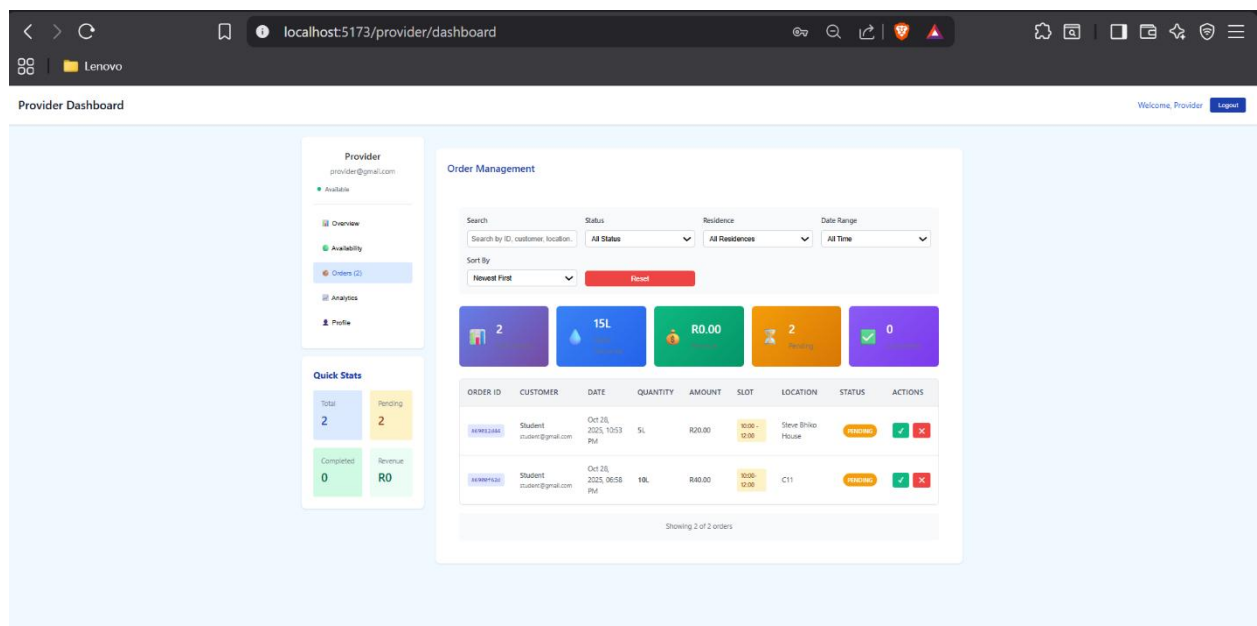


Figure 52: Provider Order Management

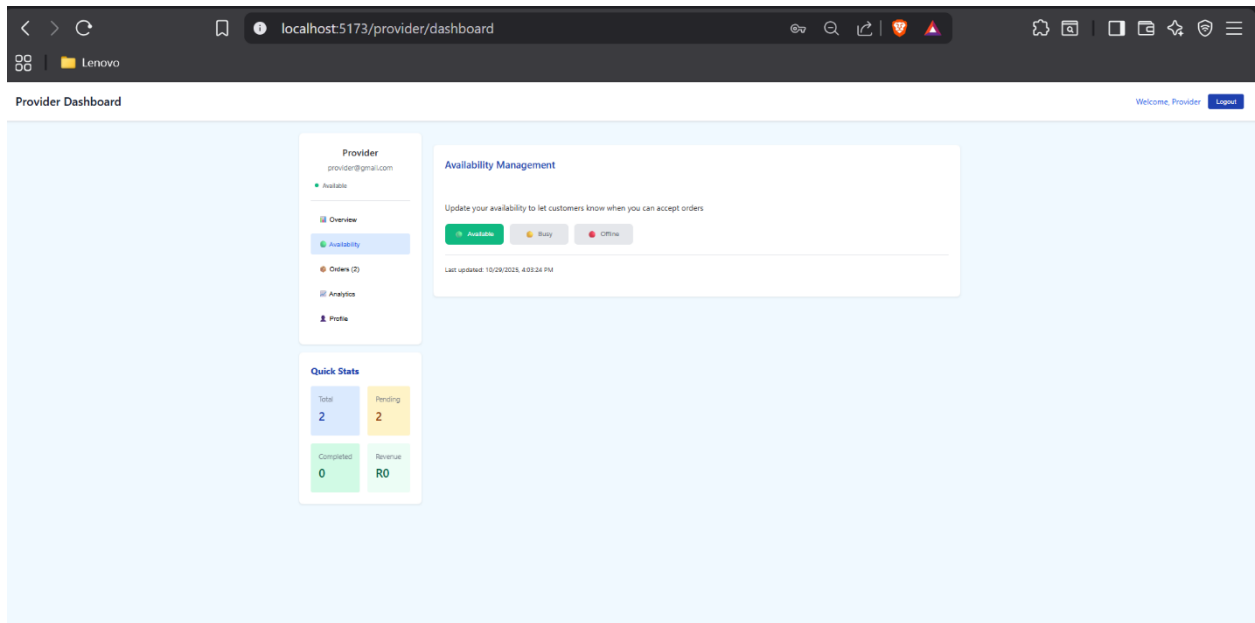


Figure 53: Provider Availability Management

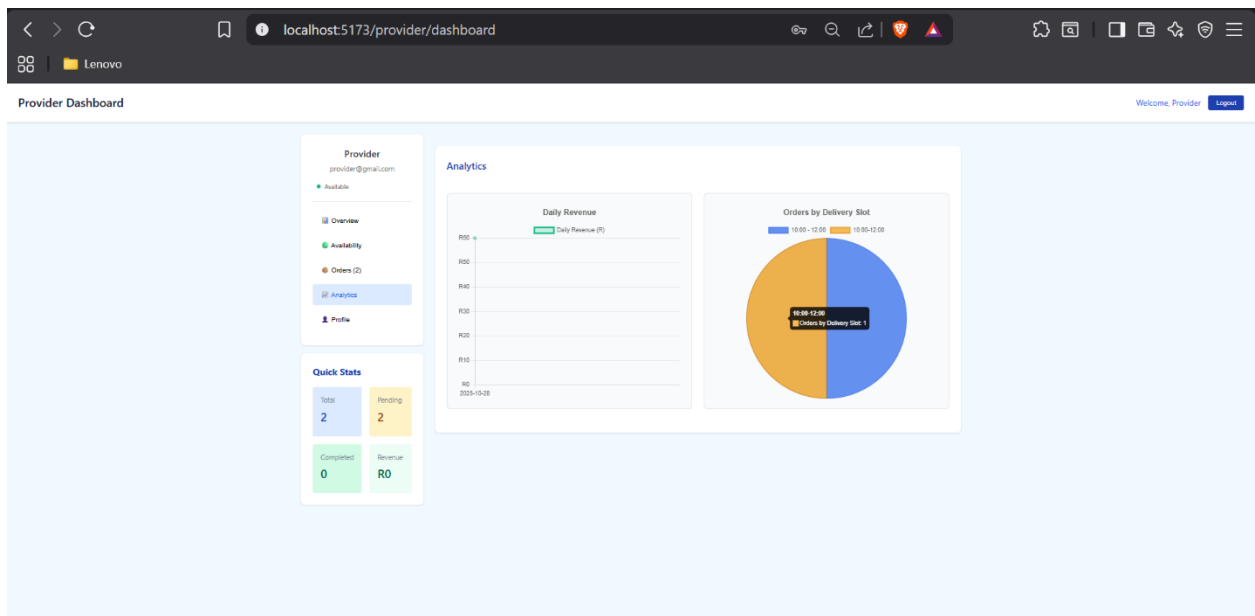


Figure 54: Provider Sales Analytics

3.3 Data Design — Logical & Physical (Normalization)

ENTITY RELATIONSHIP DIAGRAM

Amanzi ERD

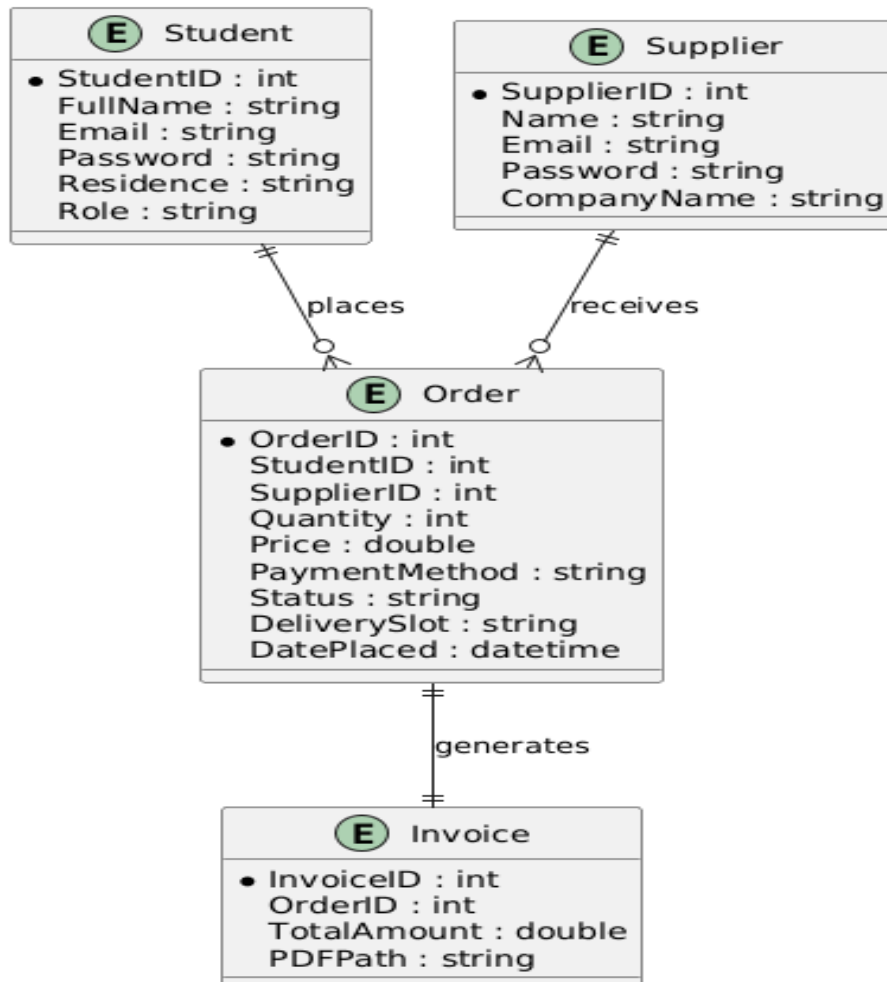


Figure 55: ERD

Main Entities and Tables (3NF)

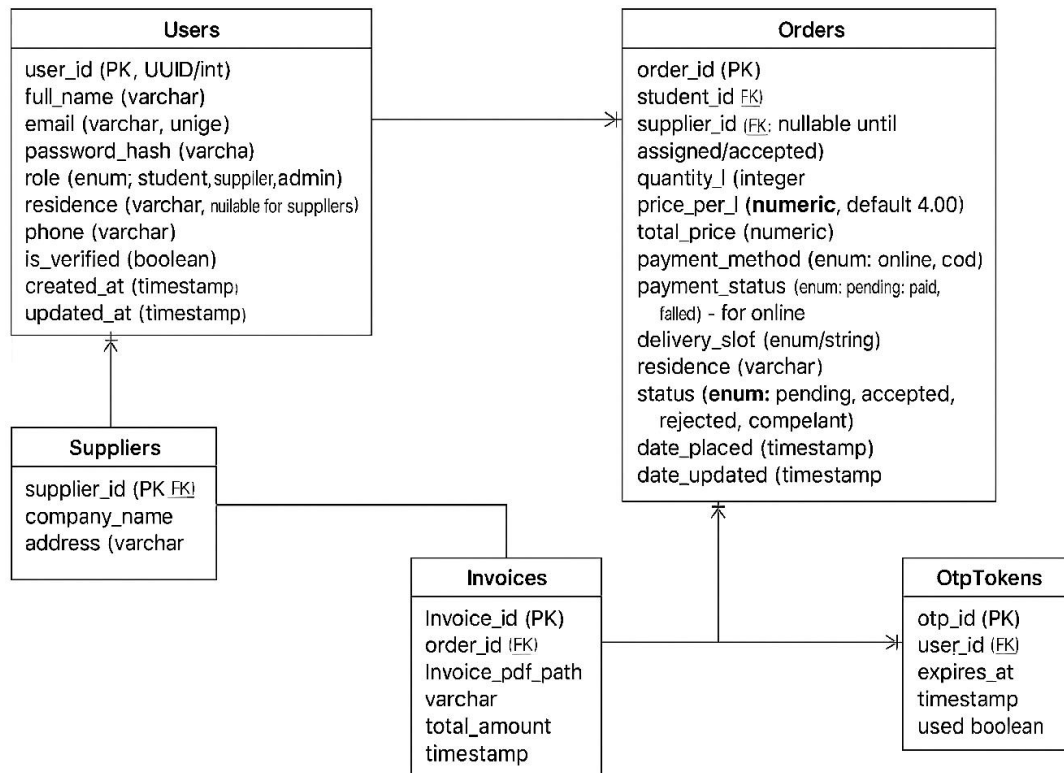


Figure 56: Class diagram

Normalization notes:

The design stores atomic data, ensures FKs for relationships, and separates invoices and OTPs from transactional orders to reduce redundancy (Shelly & Rosenblatt, 2012 — Data Design, Ch. 9).

3.4 System Architecture (Component & Deployment)

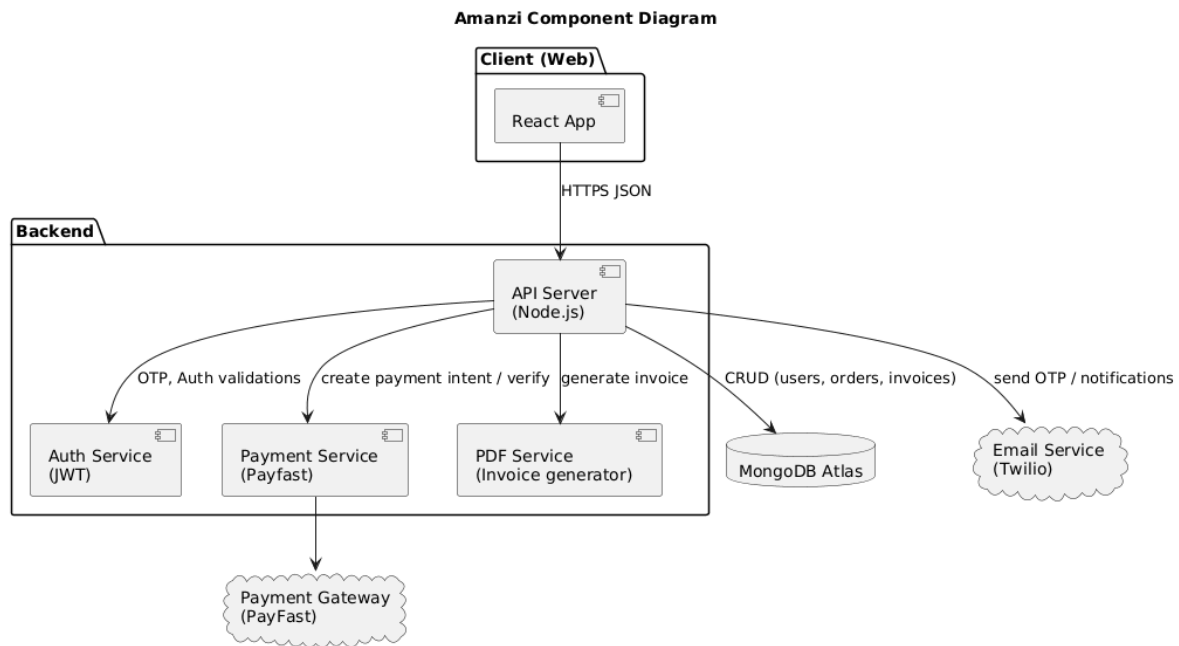


Figure 57: Component Diagram

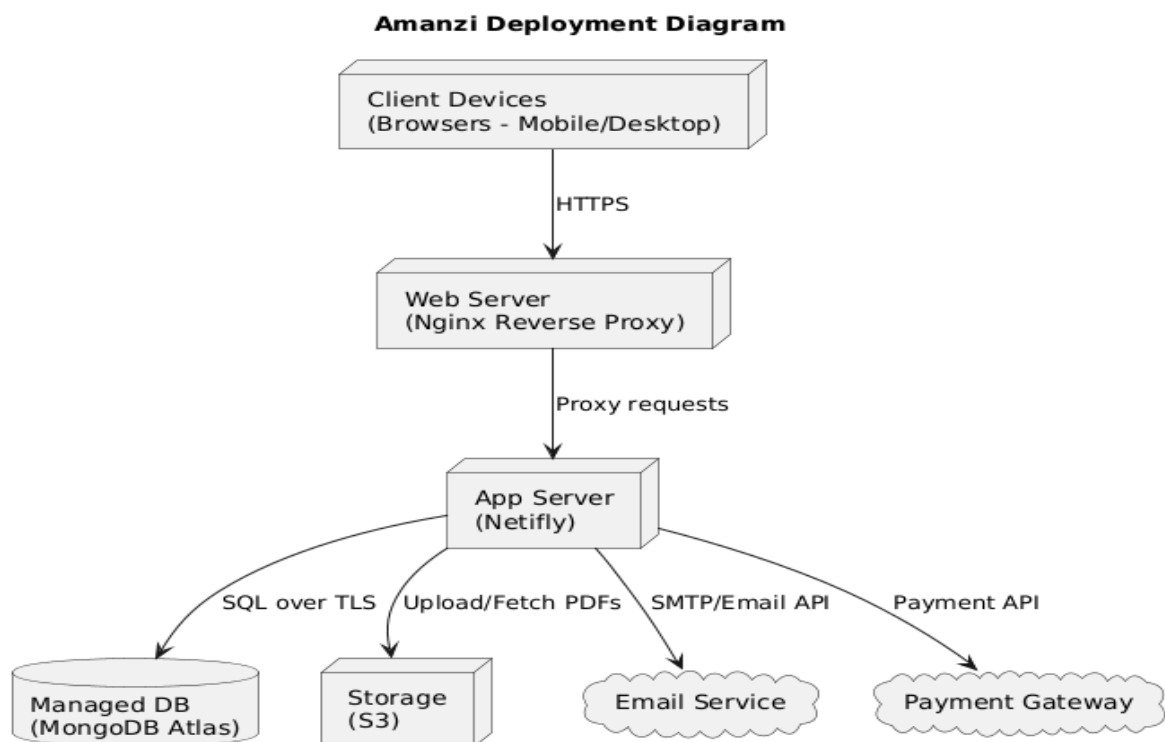


Figure 58: Deployment Diagram

Scalability considerations

- Use a load balancer + horizontal scaling of App containers.
- Cache frequently read analytics data using Redis.
- Use a CDNs for static assets.

3.5 Process & Behavior — Sequence Diagrams:

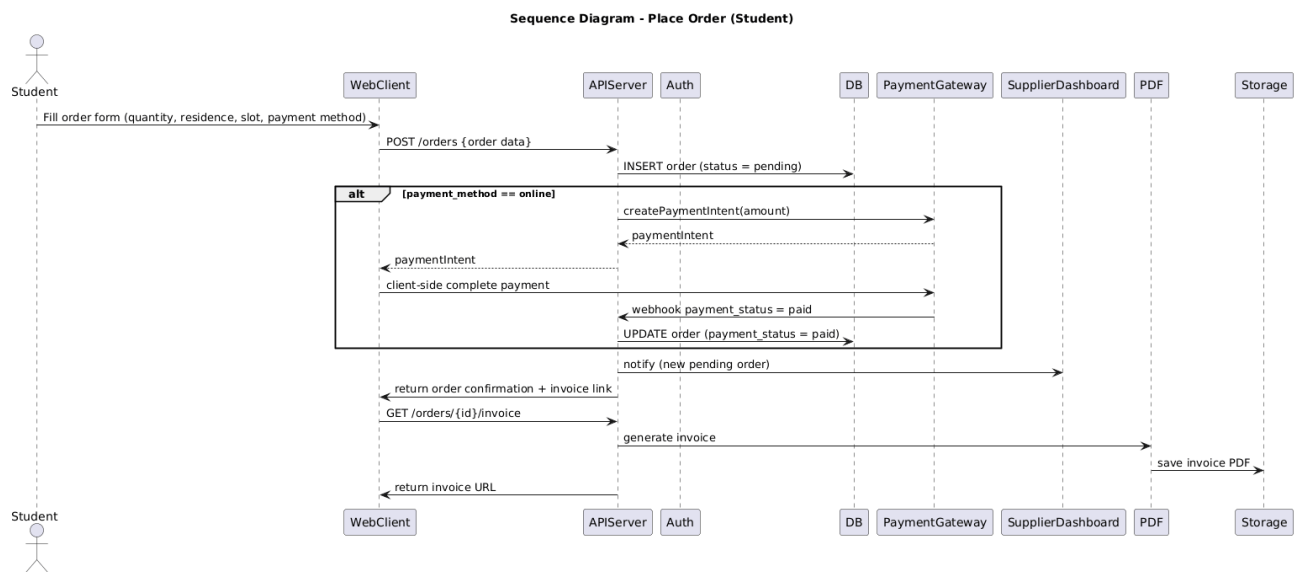


Figure 59: Place Order (Student)

3.6 API Endpoints (High-level)

- POST /api/auth/register — Register new user (role: student|supplier).
- POST /api/auth/verify-email — Verify credentials and activate account.
- POST /api/auth/login — Login (returns JWT).
- GET /api/students/me — Get student profile.
- PUT /api/students/me — Update profile.
- POST /api/orders — Place order (authenticated student).
- GET /api/orders/:id — Order details (role-based access).
- GET /api/students/orders — List student orders (filter by status).
- GET /api/suppliers/orders — Supplier order list.
- POST /api/suppliers/orders/:id/accept — Accept order.

- POST /api/suppliers/orders/:id/reject — Reject order.
- GET /api/suppliers/dashboard — Analytics: total revenue, water sold, pending.
- POST /api/payments/webhook — Payment gateway webhook.
- GET /api/orders/:id/invoice — Download invoice (auth + ownership check).

Notes:

- All endpoints must enforce role-based authorization.
- Use paginated endpoints for tables (recent orders).

3.7 Security Design

- **Authentication:** password reset via email; JWT for refresh token mechanism.
- **Password storage:** bcrypt hashing + salt.
- **Transport security:** HTTPS (TLS) for all communications.
- **Input validation:** Server-side validation for all inputs and rate limiting for endpoints like email reset and login to prevent brute-force.
- **Payment security:** Use secure payment gateway. Do not store raw card data on servers; use tokenization.
- **Role-based access control:** Ensure supplier endpoints cannot be accessed by students and vice versa.
- **Audit logging:** Record order state transitions (timestamped) for dispute resolution.
- **Backups:** Regular DB backups and S3 versioning for invoices.

(Shelly & Rosenblatt, 2012 — see System Security concepts & plan.)

3.8 Reporting & Analytics (Supplier Dashboard)

Key metrics:

- Total revenue (sum of completed orders with payment_status = paid)
- Total water sold (sum of quantity_l for completed orders)
- Pending orders count
- Orders by delivery slot (distribution)
- Recent orders table with Accept/Reject actions

Design approach

- Use aggregation queries with appropriate indexes.
- For high performance, precompute daily aggregates (cron job) and store in a supplier metrics table or cache in Redis.

3.9 Testing Plan (High-level)

Unit tests

- Auth flows, email validation, price calculation.

Integration tests

- Order placement flow with payment (simulate gateway).
- Suppliers accept/reject transitions.

End-to-end tests

- Use Cypress / Playwright to simulate a student placing an order and supplier accepting it and invoice generation.

Performance tests

- Simulate 100–1000 concurrent users placing/viewing orders (JMeter / k6).

Acceptance criteria

- All functional requirements pass in staging.
Credential match verification works; invoices downloadable; supplier dashboard reflects accepted orders; online payments processed in sandbox; role-based access enforced.

3.10 Deliverables for Implementation Phase

- Fully coded frontend and backend repo with CI/CD pipeline.
- Database migration scripts.
- API documentation (OpenAPI).
- Test suites and test reports.
- Deployment scripts.

- User manual and admin guide.

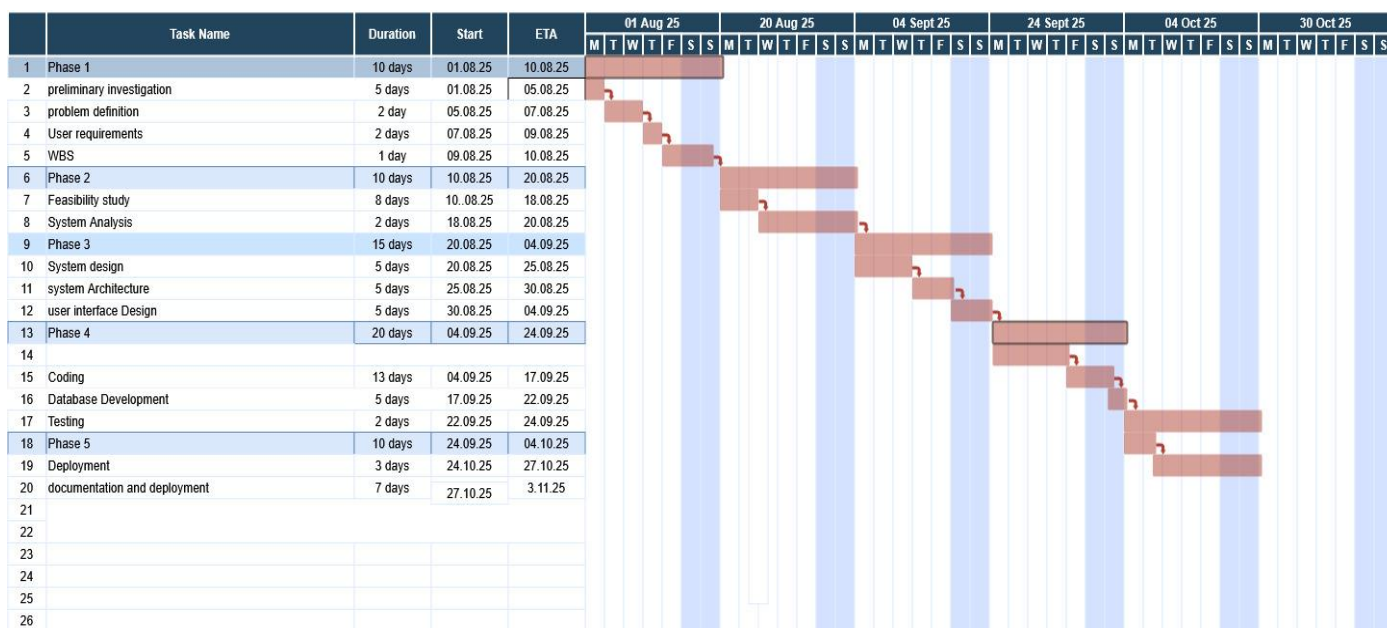
PHASE 4 – SYSTEMS IMPLEMENTATION

4 A FUNCTIONING INFORMATION SYSTEM

The goal of this is to turn the Phase 3 designs into a working, tested, and deployable Amanzi system that supports Student and Supplier roles, credentials verification, PDF invoices, and two payment methods (Online payment + Cash-on-Delivery).

4.1 Implementation Schedule — Sprints & Gantt mapping

Figure 60: Gantt chart



4.2 TEST CASES

4.2.1 Table 13- TEST CASE 1: Register new Student

Test Case # 1 System: Amanzi Ordering System Designed by: Karabo Executed by: Bheki Short description: To register new Student account	Test Case Name: Register new user (Student) Subsystem: Web application development Date: 22.09.25 Execution date: 22.09.25
---	---

Pre-Condition:

The student must sign up with the Amanzi ordering system

Step	Action	Expected Response	Pass/Fail	Comments
1.	The student enters his/her full name	Potego Sethwape	Pass	
2.	The student enters his/her email	pot123@gmail.com	Pass	
3.	The student enters his/her password.	QWERTY	Pass	
4.	The student must indicate if is student/ water supply	Student	Pass	
5.	The student selects the residence	Dr James Moroka	Pass	

Post-Condition:

New Student registered successfully.

4.2.2 Table 14 - TEST CASE 2: Log In: Student

Test Case # 2 System: Amanzi Ordering System Designed by: Bheki Executed by: Bheki Short description: To log in the Student	Test Case Name: Log in for student Subsystem: Web application development Date: 22.09.25 Execution date: 22.09.25
--	--

Pre-Condition:

The student must be signed up with the Amanzi ordering system

Step	Action	Expected Response	Pass/Fail	Comments
1.	The student enters his/her full name	Potego Sethwape	Pass	
2.	The student enters his/her email	pot123@gmail.com	Pass	
3.	The student enters his/her password.	123	Fail	The password must match

Post-Condition:

New Student registered successfully.

4.2.3 Table 15 - Test Case 3: Order

Test Case # 3 System: Amanzi Ordering System Designed by: Bheki Executed by: Bheki Short description: To order in the system	Test Case Name: Student order Subsystem: Web app development Date: 22.09.25 Execution date: 23.09.25
---	---

Pre-Condition:

The student must be logged in the Amanzi ordering system

Step	Action	Expected Response	Pass/Fail	Comments
1.	The student selects water quality(litres) to order	5	Pass	
2.	The student selects Residence	Dr James Moroka	Pass	
3.	The student chooses delivery slot	6 PM	Pass	
4.4.	The student places order	Press place order button	Pass	

Post-Condition:

New Student registered successfully.

4.2.4. Table 16 - Test Case 4: View Status

Test Case # 4 System: Amanzi Ordering System Designed by: Potego Executed by: Bheki Short description: To order in the system	Test Case Name: Student order Subsystem: Web application development Date: 23 . 09 . 25 Execution date: 23 . 09 . 25
--	---

Pre-Condition:

The student must be logged in the Amanzi ordering system

Step	Action	Expected Response	Pass/Fail	Comments
1.	The student can track their orders	Track	Pass	

Post – Condition

The student must be logged in

4.2.5. Table 17 - Test Case 5: View Order history

Test Case # 5 System: Amanzi Ordering System Designed by: Potego Executed by: Bheki Short description: Student view their order history	Test Case Name: view order analytics Subsystem: Web application development Date: 23 . 09 . 25 Execution date: 23 . 09 . 25
--	--

Pre- Condition:

The be logged in the system

Step	Action	Expected Response	Pass/Fail	Comments
1.	View the order analytics	Track order history on data history	Pass	

Post - Condition:

The order history shows order analytics

Test Cases for the service providers

Table 18 - TEST CASE 6: Register new Service provider

Test Case # 6 System: Amanzi Ordering System Designed by: Karabo Executed by: Bheki Short description: To register new service provider	Test Case Name: Register new service provider Subsystem: Web application development Date: 23 . 09 . 25 Execution date: 23 . 09 . 25
--	---

Pre-Condition:

The student must sign up with the Amanzi ordering system

Step	Action	Expected Response	Pass/Fail	Comments
1.	The student enters his/her full name	Karabo Makau	Pass	
2.	The student enters his/her email	makaukarabo538@gmail.com	Pass	
3.	The student enters his/her password.	QWERTY12	Pass	
4.	The student must indicate if is student/ water supply	Service provider	Pass	

Post-Condition:

New service provider registered successfully.

Table 19 - Test case 7: Log in

Test Case # 7 System: Amanzi Ordering System Designed by: Bheki Executed by: Bheki Short description: To log in the service provider	Test Case Name: Log in for service provider Subsystem: Web application development Date: 23.09.25 Execution date: 24.09.25
---	---

Pre-Condition:

The service provider must be signed up with the Amanzi ordering system

Step	Action	Expected Response	Pass/Fail	Comments
1.	The student enters his/her full name	Karabo Makau	Pass	
2.	The student enters his/her email	makaukarabo538@gmail.com	Pass	
3.	The student enters his/her password.	xxxxxxxxxxxxxx	Fail	The password must match

Post-Condition:

New Service provider registered successfully.

Table 20 - Test Case 8: View Order history

Test Case # 8 System: Amanzi Ordering System Designed by: Bheki Executed by: Bheki Short description: Service provider accept/decline orders	Test Case Name: view order Subsystem: Web application development Date: 24.09.25 Execution date: 24.05.25
---	--

Pre- Condition:

The be logged in the system

Step	Action	Expected Response	Pass/Fail	Comments
1.	Service Provider can view orders	Service can accept or decline the orders	Pass	

Post - Condition:

The service provider accepts and declines the orders

Table 21 - Test Case 9: view order analytics

Test Case # 9 System: Amanzi Ordering System	Test Case Name: view order analytics Subsystem: Web application development Date: 24.09.25
---	---

Designed by: Bheki Executed by: Bheki Short description: Service provider views order analytics	Execution date: 24.09.25
--	---------------------------------

Pre- Condition:

The be logged in the system

Step	Action	Expected Response	Pass/Fail	Comments
1.	View the order analytics	Track orders history on data history	Pass	

Post - Condition:

The order history shows order analytics

4.3 SYSTEM INSTALLATION/CONVERSION STRATEGY:

The Amanzi Purified Water Ordering and Delivery Information System will be introduced using a Pilot Conversion Strategy, as recommended by Shelly & Rosenblatt (2012), for systems that require controlled testing, user feedback, and low-risk deployment. In a pilot approach, the new system is implemented in a single, limited area within the organization before full campus-wide rollout. This ensures that potential issues can be identified and resolved early without affecting the entire user population.

For Amanzi, the3 pilot site will be:

4.3.1 Selected Pilot Group:

- NWU Mahikeng Campus – Cluster Residences.

Cluster Residences have been selected because they:

- Have a high population of active student residents using delivery services.
- Provide a manageable sample size for observing system performance.
- Allow the project team to closely monitor student ordering behaviour, supplier response times, and delivery operations.
- Offer an appropriate environment for usability testing and refinement before scaling across campus.

4.3.2 Objectives of the Pilot

The pilot phase aims to:

1. **Validate system functionality** in a real operational environment (registration, Credentials email verification, ordering, payment, supplier dashboard, etc.).
2. **Identify defects or usability issues** before full deployment.
3. **Evaluate system performance**, responsiveness, and reliability under actual user load.
4. **Assess supplier workflow**, including order acceptance/rejection, delivery management, and analytics.
5. **Gather feedback** from students and suppliers to improve UX, interface clarity and business rules.
6. **Mitigate risk** by limiting exposure of potential failures to a small group.

4.3.3 Pilot Duration

- **Initial Pilot Runtime:** 1-2 weeks
- Continuous monitoring will occur during this period.
- Extension may occur if additional testing is required.

4.3.4 Pilot Activities

During the pilot conversion:

- Students in Cluster Residences will register, verify accounts, place orders, and make payments (COD and online).
- Supplier partners will accept or reject orders and view analytics via the supplier dashboard.

The project team will track:

- Order volumes
- Payment success rate
- Delivery times
- Dashboard accuracy
- Error logs
- User experience feedback

Regular meetings will be held with pilot participants to capture feedback and report issues.

4.3.5 Success Criteria for the Pilot

The pilot will be considered successful once the following conditions are met:

- Login Credentials verification works reliably for first-time users.
- Forgot/Reset Password works for those who have forgotten their passwords.
- Students can place orders with accurate pricing.
- Orders flow correctly from **Pending > Accepted/Rejected > Completed**.
- Invoices are generated automatically and downloadable.
- Supplier dashboard metrics (revenue, total litres sold, pending orders) are calculated correctly.
- Online payments are processed successfully through the integrated payment gateway.

- No critical bugs remain unresolved.
- User satisfaction feedback from pilot participants => 80%.

4.3.6 Post-Pilot Actions

Once the pilot is completed:

1. **Collect and review pilot feedback** from both students and suppliers.
2. **Fix defects**, apply updates, and run regression tests.
3. **Optimize system performance** based on observed usage metrics.
4. **Receive final approval from project stakeholders.**
5. **Roll out Amanzi to the entire NWU Mahikeng Campus using the lessons learned from the pilot.**

4.3.7 Rationale for Choosing Pilot Conversion

The pilot approach is suitable for Amanzi because:

- The system introduces **new digital processes** (ordering, delivery schedule tracking, credentials verification, online payments) that have not been previously used by students or suppliers at Mahikeng for ordering water traditionally.
- A full direct cutover poses high risk if critical issues occur.
- Pilot conversion reduces organizational risk while enabling real-world testing.
- Feedback from a controlled environment enhances system quality before full-scale deployment.

According to Shelly & Rosenblatt (2012), **pilot conversion** is ideal when organizations require “a safe and manageable environment to validate the new system before enterprise-wide adoption.”

PHASE 5 – SYSTEMS SUPPORT AND SECURITY

5. AN OPERATIONAL SYSTEM THAT IS PROPERLY MAINTAINED, SUPPORTED, AND SECURED

5.1 Introduction

The Amanzi Water Ordering Information System enters its support and security phase after successful deployment and implementation. This phase focuses on maintaining smooth operations, enhancing performance, and ensuring system security. Continuous monitoring, user support, and proactive maintenance are essential to sustain system reliability and efficiency. The goal is to keep the system aligned with user needs, technological advancements, and institutional policies at North-West University.

5.2 User Support

The system's success depends on ongoing user satisfaction and proper support structures.

User support for the Amanzi system includes:

- **User Training:** New users (students and service providers) receive orientation through online help documentation and step-by-step tutorials within the system dashboard. This helps them understand order placement, payment procedures, and dashboard usage.
- **Help Desk and Feedback System:** A built-in support feature allows users to report technical problems or request assistance. Support tickets are logged and reviewed by the system administrator team for resolution.
- **Outsourced Technical Assistance (if required):** While the system is developed and managed internally, future maintenance or updates may be outsourced to certified software professionals under NWU's IT policy for efficiency and specialization.

5.3 Maintenance Tasks:

Maintenance ensures the Amanzi system continues to meet operational, technical, and security requirements. Based on Shelly & Rosenblatt (2012), the maintenance process includes four categories:

- **Corrective Maintenance:** Addresses bugs or issues identified during operation. For example, correcting database connection errors, fixing payment gateway malfunctions, or resolving login authentication bugs.
- **Adaptive Maintenance:** Implements changes due to new requirements or external conditions. For instance, if the NWU IT department updates its data security policy or if Stripe changes its API standards, the system's backend code will be adapted accordingly.
- **Perfective Maintenance:** Focuses on improving performance and usability. Enhancements may include faster data loading, improved mobile responsiveness, or redesigned dashboards for better user experience.
- **Preventive Maintenance:** Aims to avoid potential problems by performing regular updates and system audits. Examples include scheduled backups, API updates, and security patches to prevent vulnerabilities.

5.4 Maintenance Management:

Effective maintenance management ensures structured updates and version control for Amanzi.

Key aspects include:

- **Maintenance Team:** The project manager, backend developer, and database administrator collaborate to manage upgrades, troubleshoot errors, and maintain documentation.
- **Configuration Management:** All system modifications are logged through version control tools such as GitHub, maintaining a clear record of changes to frontend and backend components.
- **Release and Version Control:** System changes are compiled into periodic updates (e.g., Amanzi v1.1, v1.2) rather than ad-hoc fixes, ensuring stability.
- **Baselines:** Performance and configuration baselines are defined to measure and compare future enhancements objectively.

5.5 System Performance Management:

Regular performance monitoring ensures system reliability.

For Amanzi, performance metrics include:

- **Response Time:** Pages and dashboards should load within 2–3 seconds.

- **Uptime:** System availability should remain above 99%.
- **Capacity Planning:** Database scalability ensures handling of 500+ concurrent student logins and orders.
- **Fault Management:** Logs are automatically generated for errors, helping administrators detect, report, and resolve issues quickly.

5.6 System Security:

Security remains a top priority to protect user data, transactions, and system operations.

Following Shelly & Rosenblatt's six levels of security, Amanzi incorporates:

- **Physical Security:** Hosting servers on secure cloud infrastructure (MongoDB Atlas, Netlify) with restricted administrative access.
- **Network Security:** All communications are encrypted using SSL/TLS, protecting data between client and server.
- **Application Security:** Regular vulnerability testing is performed to identify risks in login modules, API endpoints, and payment gateways.
- **File Security:** Uploaded documents and stored data are restricted to authorized roles, ensuring compliance with the POPI Act.
- **User Security:** Login authentication uses hashed passwords and verification emails for password recovery, preventing unauthorized access.
- **Procedural Security:** Administrative procedures such as periodic audits, access control reviews, and user training enforce safe data practices.

5.7 Risk Management:

Risk management identifies and mitigates potential threats to system stability.

For the Amanzi system:

- **Risk Assessment:** Regular reviews check for vulnerabilities in API integrations, payment handling, and authentication.
- **Mitigation:** Backups, secure passwords, and firewall protections are enforced.
- **Incident Response:** A contingency plan outlines procedures for data breaches or service downtime, ensuring minimal disruption.

5.5 System Performance Management:

Regular performance monitoring ensures system reliability.

For Amanzi, performance metrics include:

- **Response Time:** Pages and dashboards should load within 2–3 seconds.
- **Uptime:** System availability should remain above 99%.
- **Capacity Planning:** Database scalability ensures handling of 50+ concurrent student logins and orders.
- **Fault Management:** Logs are automatically generated for errors, helping administrators detect, report, and resolve issues quickly.

5.9 System Obsolescence:

Over time, the Amanzi system may require major upgrades or replacement due to:

- Technological advancements in web development frameworks.
- Shifts in NWU's IT infrastructure or policies.
- Increased user demand beyond system capacity.

When the system becomes inefficient to maintain, a new development cycle will begin, initiating Phase 1 – Systems Planning of a new SDLC iteration.

5.10 Summary

The **Amanzi Water Ordering Information System** is designed for sustainability and security through proactive support and maintenance. This phase ensures that the system remains functional, secure, and efficient throughout its operational life. By applying structured maintenance, performance monitoring, and layered security measures, Amanzi continues to provide a reliable, user-centered, and secure digital platform for water ordering and delivery at North-West University.

BIBLIOGRAPHY

1. Journal Article:

Abu-Alhaija, A.S. 2019. From epistemology to Structural Equation Modeling: An essential guide in understanding the principles of research philosophy in Selecting the Appropriate Methodology. *Australian Journal of Basic and Applied Sciences*, 13(9):122-128.

2. Online Research:

Barragán-Landy, M., Sousa, S., Romero, F. & Leão, C. 2020. A Proposed Representative Sampling Methodology. In. 20th European Conference on Research Methodology for Business and Management Studies: ECRM. p 8.

3. Textbook:

Shelly, G.B. & Rosenblatt, H.J. 2012. *Systems Analysis and Design*. 9th ed. Boston, MA: Course Technology, Cengage Learning.

5. ANNEXURE A: DATA COLLECTION TOOL



ANNEXURE B: SYSTEM PERFORMANCE EVALUATION TOOL



ANNEXURE C: SYSTEM SECURITY AUDIT TOOL

